

Watch: European pharma's fight to stay relevant

How does Europe need to do to stop its pharmaceutical industry from becoming 'yesterday's news'? ING's Diederik Stadig says the sector still has trump cards to play, but it needs to act fast to avoid being squeezed out by US and Chinese competition.

Read more about Diederik's thinking [here](#)



European pharma's fight to stay relevant

Europe's pharmaceutical industry needs to up its game if it's going to continue competing effectively with the likes of China and the US. So says ING's Senior Economist for Healthcare and Technology, Diederik Stadig. He points out that in 1990, nearly half of all global Research & Development spending took place in Europe. Today, it's just 26%. And he outlines what needs to happen to stop the European pharma sector from being 'yesterday's news'.

[Watch video](#)

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Norges Bank is ready to hike rates

Norway's central bank has signalled a hike is coming soon. Its inflation concerns are broad-based, not just limited to the energy price shock, and we now expect a hike in May – although timing depends on war developments. With 60bp priced in this year, we think this is the peak for front-end NOK rates, but our baseline EUR/NOK profile remains downward sloping



We're currently calling for one rate hike on the back of our bearish baseline view for oil and gas prices, but the chances of two (fully priced in) are elevated

Hawkish shift beyond expectations

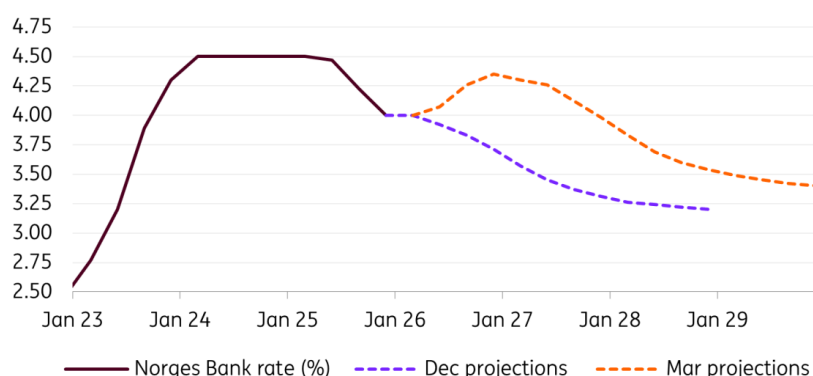
We had expected a hawkish shift in today's Norges bank meeting, but the kind of commitment to rate increases that emerged in the statement came as a surprise. A near 10bp jump in 1-year NOK swap rates – on top of the +43bp move since the start of the war – signals markets were equally surprised by the hawkish messaging.

Norges Bank kept rates at 4.0% today, but said “it will likely be appropriate to raise the policy rate at one of the forthcoming monetary policy meetings.” The updated rate projections now fully embed a 25bp rate hike in the third quarter, and a roughly 50% probability of a further 25bp increase in the fourth quarter. That is a significant shift of direction from December's Monetary Policy Report, which still signalled one rate cut in 2026.

Inflation projections have been revised 1ppt higher to 3.4% for 2026, and core CPI 0.6ppt higher to

3.3%. Growth for 2026 is still expected at 1.2% though, with downward revisions for 2027 (-0.2 ppt) and 2028 (-0.6ppt).

Major upward revisions in rate projections



Source: Norges Bank, ING

Energy price shock part of bigger inflation problem

The details of the members' discussions – published for the first time – further reinforce the hawkish message. Alongside the numerous warnings for elevated uncertainty warranting caution, it emerged that “some” members were ready to raise interest rates. The Monetary Policy Committee is made of only five members, suggesting this was a much closer decision than markets had anticipated, with only 6bp priced in.

Even more crucially, the committee's concerns about inflation appear much broader than the energy price shock. Some members highlighted how inflation had remained above target for some time, and the recent stickiness was not due to temporary factors but rather is broad-based. The acceleration in rent inflation is a case in point.

Wage growth remains the other key worry. The Norwegian Technical Calculation Committee for Wage Settlements projected inflation at 3.2% in 2026, well above the Norges Bank's December estimate of 2.5%. The concern here is that the wage environment, alongside low unemployment, can offer a breeding ground for inflation to entrench.

A hike almost inevitable

Our macro team has been positioning on the dovish side for most major central banks relative to the market pricing, still expecting no hikes by the Federal Reserve, the European Central Bank and the Bank of England. We would have been reticent to pencil in Norges Bank hikes if the Committee had only pointed to energy prices as a reason for their hawkish shift.

But explicit concerns about broad-based inflationary issues and indications that other macro factors can favour inflation entrenching in Norway make us believe at least one hike is now looking likely. Incidentally, the committee noted that should it not hike, the krone could depreciate and remove a cushion for reduced imported inflation.

Oil prices and incoming inflation figures will determine the timing. Markets are pricing in 16bp for May and 33bp for June. Considering some members already wanted to hike today, May looks

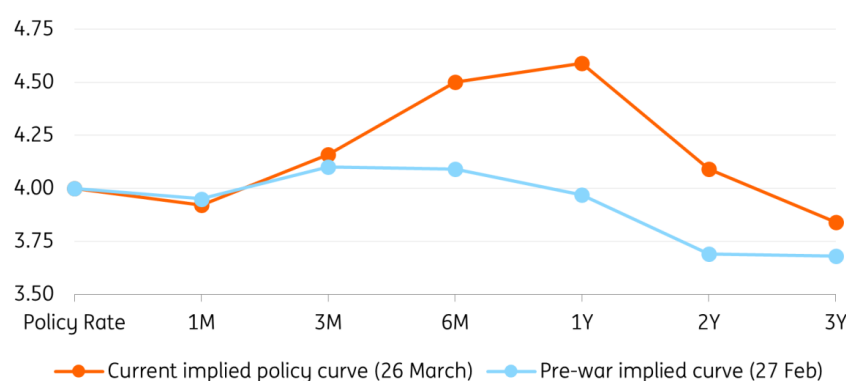
slightly more likely. For now, we are only calling for one rate hike on the back of our [bearish baseline view for oil and gas prices](#), but the chances of delivering two (fully priced in) are elevated.

The bar for a dovish re-think is high

Can Norges Bank back away from hikes? Yes, after all, it has a recent history of committing to a policy move and not delivering. In January 2025, it stated that “the policy rate will likely be reduced in March” but then reconsidered as inflation rose.

In this case, though, we need to be clear on the scenario that could prompt a dovish rethink: a sharp decline in oil and gas prices associated with a rapid de-escalation and re-opening of the Strait of Hormuz. Core inflation would also need to show limited upside and the decline in the krone would need to prove relatively contained – perhaps on the back of a major equities rebound.

NOK swaps fully embedding two hikes



Source: Refinitiv, ING

Front-end NOK rates close to peak

The NOK swap curve currently shows 60bp of tightening in the next year. As discussed above, two hikes are a tangible possibility (even if not our base case for now), but the 4.0% policy rate is already restrictive, especially in real terms. Even in Norges Bank’s latest projections, inflation would peak around 3.5%.

Incidentally, some members voiced concerns about the growth and employment risks of overtightening. We feel we are very close to the peak for front-end NOK swap rates and see risks on the downside.

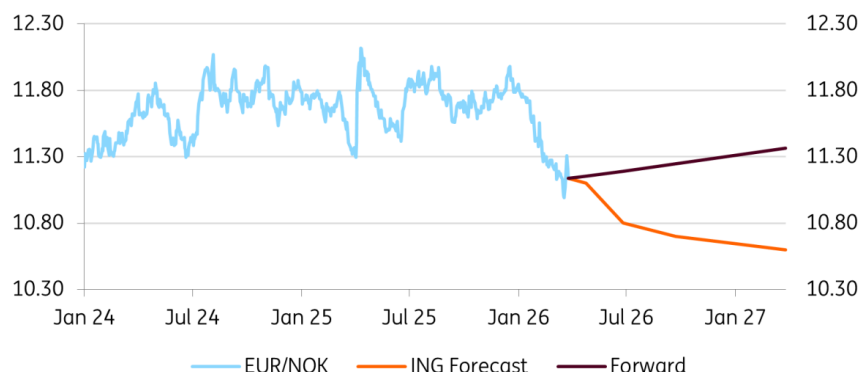
NOK outlook remains solid

The implications for NOK are more nuanced. A major de-escalation and drop in energy prices could harm NOK, but much will depend on where markets see oil and gas prices stabilising beyond the initial knee-jerk reaction.

A scenario where Brent oil settles around 80-90\$/bbl and TTF gas around 40-50€/MWh into the summer (our baseline) could be enough to generate a positive risk sentiment response while keeping markets attracted to commodity currencies like NOK as well as Norges Bank on track for a hike. That is a scenario where EUR/NOK can make a decisive break below 11.0 in the near term.

For now, we are sticking with our EUR/NOK forecast of 10.80 for the second quarter, but risks are on the downside due to potential widening in Norges Bank-ECB rate expectations. Our year-end target is 10.60.

Our EUR/NOK forecast



Source: ING, Refinitiv

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French businesses and consumers are becoming more pessimistic

While the March surveys only imperfectly capture the effects of the conflict in the Middle East, the signals sent by French businesses and households are already clearly pointing downwards



The business climate in France is below its long-term average

Sentiment is deteriorating

The business climate in France remained stable in March, at 97, but has now been below its long term average for almost two years. Responses were collected by INSEE between 26 February and 23 March. However, the vast majority of firms responded before mid March, at a time when the repercussions of the conflict in the Middle East were still limited compared with the current situation. Nearly one fifth of responses date from before the outbreak of the conflict.

As expected, the business climate deteriorated sharply in industry, losing three points over the month. Past production, production expectations and order books are all down. This deterioration is mainly driven by the “other transport equipment” sector, which had been one of the main engines of French production in recent months but is now markedly more pessimistic. At the same time, selling price expectations in industry rose sharply over the month.

Sentiment remains very depressed in services, due to a more negative assessment of past activity. By contrast, the business climate edged up slightly in retail trade and construction. The

employment climate improved by one point, but remains well below its long term average.

Household confidence also declined in March, falling by two points over the month. Although only half of survey responses were collected after the outbreak of the conflict in the Middle East, households are anticipating a very sharp rise in prices in the coming months. The corresponding balance of opinion jumped by 28 points and reached its highest level since September 2022. Households are also markedly more pessimistic about the future evolution of their personal financial situation.

Overall, the data published today confirm that French economic prospects have deteriorated significantly since the outbreak of the conflict in the Middle East. Given the survey collection timetable, the impact of the most recent developments is not yet fully reflected in the March data. If the conflict does not ease in the coming days, a much sharper deterioration in sentiment can be expected in April.

French inflation will rise, but less than elsewhere in Europe

The impact of the situation in the Middle East on the French economy will largely depend on the duration and intensity of the conflict. One consequence already appears certain, however: inflation will be higher than previously expected. At the start of 2026, our forecasts pointed to average inflation of 1.3% for the harmonised consumer price index in France. We now expect inflation to be close to 2% over the year as a whole, with a risk of further upward revisions.

That said, this revision remains more limited than in other European countries. France's inflation starting point is lower than in most other euro area countries. In addition, the French economy is characterised by a large share of administered or centrally negotiated prices, notably in large scale retail and energy. This means that increases in commodity prices are transmitted more slowly to consumer prices than elsewhere. Finally, the economic environment prior to the conflict was not particularly dynamic, which reduces the risk of a rapid inflationary spiral.

The labour market also remains fragile. As shown by the employment climate, hiring intentions are weak and net job creation has been negative for several quarters. Wage negotiations for this year do not incorporate this new inflationary environment. As a result, wage growth should remain moderate, helping to contain price increases, particularly in services.

The drivers of economic activity will come under strain

The risk of a rapid inflationary spiral is therefore more limited in France than in neighbouring countries, but it is not negligible in the event of a prolonged conflict. Moreover, the risk of a negative impact on economic activity is more worrying in France than elsewhere.

The French economy had started 2026 with a slight improvement in optimism, but on fragile foundations. Confidence had only just begun to recover after a difficult 2025, marked by strong political and fiscal uncertainty that weighed on consumption and investment decisions. The current situation is reviving these uncertainties, which should slow the decline in the household savings rate and limit hiring decisions. Rising energy prices will also weigh on purchasing power. An acceleration in consumption now seems unlikely.

In addition, a rebound in investment had been expected at the start of the year, both among households and businesses. However, the rapid rise in long term interest rates since early March,

linked to expectations of monetary tightening by the European Central Bank, is now jeopardising this scenario. Investment growth is therefore likely to remain subdued in 2026. Finally, it should be remembered that the dynamism of French production relies heavily on the aerospace sector. This sector is particularly exposed to the current situation, raising fears of a decline in orders in 2026.

As a result, even if the conflict were to ease in the coming weeks, the main drivers of economic activity already appear weakened. This should translate into weaker than expected growth in 2026. At this stage, we expect average growth of 0.7% over the year as a whole, compared with 1% anticipated at the start of the year. Risks surrounding this forecast remain tilted to the downside: a prolonged conflict could lead to an even more pronounced downward revision of growth prospects.

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FX Daily: USD needs a spark for the next leg lower

Risk sentiment has improved, but oil prices sticking above \$100/bbl are keeping USD afloat. Markets may well require some more convincing headlines on de-escalation to take the dollar meaningfully lower from here. In Norway, we expect a rate hold this morning, but a hawkish shift seems likely



Oil prices staying above \$100/bbl are keeping the dollar afloat

➔ USD: Finding a firmer footing

Risk sentiment has improved over the past 24 hours thanks to some constructive headlines on the Middle East conflict. Iran continues to deny that ceasefire talks with the US are ongoing, but many could be interpreting this as a way to maintain the leverage of high oil prices for as long as possible.

Still, the FX market isn't ready to add another leg of de-escalation trade just yet. After all, oil prices still above \$100/bbl argue against aggressive USD selling, and the dollar is showing signs of better resilience after Monday's selloff. Some confirmation from Iran on the ongoing peace talks remains necessary to see DXY trade back below 99.0, in our view.

On the Fed side, we've heard a few hawkish leaning comments this week, but with little in the way

of incremental guidance for markets. That confirms our expectation that the FOMC communication impact would wear off quickly – oil prices now remain the only clear driver of rate expectations. We'll hear from Cook, Miran, Jefferson and Barr today, while the US data calendar remains quiet, with only jobless claims in focus today.

Francesco Pesole

➔ EUR: ECB pricing remains volatile

EUR/USD eased back below 1.16 yesterday despite good risk sentiment, which could be mirroring some profit-taking on the de-escalation trade. At the same time, EUR/USD did jump on the ECB's surprise hawkishness last week, and may be giving up some of those gains as the dovish repricing in ECB expectations proves steeper than for the Fed when oil prices decline.

Markets have notably scaled back expectations for an April ECB hike, which was priced at a peak 22bp on Monday and is now at 17bp after having fallen to 14bp – closely tracking oil prices.

Yesterday, ECB President Christine Lagarde said the central bank wouldn't be paralysed by hesitation, but we also heard some slight dissent from dovish member Villeroy, who stressed it's too early to discuss the timing of hikes.

Any extra day without tangible de-escalation in the Gulf dents the sustainability of the latest EURUSD rally and argues for a return below 1.150. No key data to watch in the eurozone today, but the ECB's Guindos and Muller are due to speak.

Francesco Pesole

➔ GBP: BoE speakers in focus

We'll hear from three Bank of England MPC members today: Sarah Breeden, Alan Taylor and Megan Greene. Greene already spoke yesterday and reiterated her inflation concerns and hawkish spin. Breeden claimed she would have voted for a cut at last week's meeting if it wasn't for the energy price spikes, but was generally hawkish too. Let's see if signs of potential de-escalation make her turn slightly more dovish.

Alan Taylor was the only dove who stayed dovish last week, arguing that the implications for medium-term inflation from temporary energy shocks aren't significant and the bar for hikes is high. But more dovish comments from him won't come as a surprise to markets.

We still see some upside risks for EUR/GBP due to the larger room for dovish repricing in a de-escalation scenario for the GBP curve. A move past 0.870 in the coming weeks remains our baseline.

Francesco Pesole

➔ NOK: Norges Bank to deliver its hawkish shift

Norges Bank announces its monetary policy this morning. A hold at 4.0% looks very likely, with only a marginal chance of a hike (around 6bp priced). The new rate projections will be the main focus. The December update still pointed to a rate cut in 2H26, but it seems likely they will revise to show a flat profile at 4.0% or some tightening.

Another new feature of this meeting is the publication of a summary of opinions alongside the statement and the Monetary Policy Report. Recent central bank meetings suggest that hawkish surprises have tended to emerge more from individual members' comments than from the otherwise cautious official statements.

Markets are currently pricing around 25bp of cuts by June, a theme that – as elsewhere – is tightly linked to energy prices. However, Norges Bank's policy is already restrictive and underlying inflation had eased before the war, with core CPI back to around 3.0% in February.

While some hawkish messaging today could keep a June hike notionally on the table, our broader view on Norges Bank remains similar to that of other European central banks. Current pricing still looks too aggressive, with risks skewed clearly to the dovish side.

We retain a short-term 11.10 target in EUR/NOK. A scenario of de escalation, combined with a rebound in equities and only a gradual retracement in energy prices, would be ideal for the krone and could allow EUR/NOK to break below 11.00 before the summer. For now, we keep our end 2Q target at 10.80.

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The Commodities Feed: Oil volatile as ceasefire talks stall

Oil prices are up around 2% this morning as conflicting US Iran signals on a potential ceasefire keep the geopolitical risk premium elevated



Energy – Oil volatile as ceasefire talks stall

NYMEX WTI and ICE Brent rose around 2% this morning amid conflicting statements from the US and Iran on the prospects of a ceasefire.

Oil prices were volatile on Wednesday, trading on either side of \$100/bbl, as markets weighed mixed signals around potential US Iran talks. Iran rejected a 15 point US truce proposal, which included limits on its nuclear and missile programmes, and set out its own conditions for ending the war, including security guarantees, compensation for damages and recognition of its authority over the Strait of Hormuz.

Tehran has continued its attacks on Israel and Arab Gulf states, while vessels seeking to transit the Strait under Iranian protection are reportedly being asked to provide details of their crew and cargo. Meanwhile, Iran's Bushehr nuclear power plant was reportedly attacked again, following an earlier strike in mid March that the IAEA said caused no damage to the reactor.

Separately, US Vice President JD Vance may reportedly travel to Pakistan for Iran talks this

weekend.

Any credible de escalation could trigger a renewed risk on move, but for now uncertainty remains elevated.

US oil inventories continued to build, with EIA data showing crude stocks rising by 6.9m barrels last week – marking a fifth consecutive increase and well above the 2.3m barrel build flagged by the API. Total crude stocks climbed to 456.2m barrels, the highest since June 2024, while Cushing inventories surged by 3.4m barrels to 30.9m barrels, the biggest weekly gain since January 2023. Crude imports fell to 6.5m b/d, while exports dropped sharply to 3.3m b/d, the lowest since November 2025.

Product balances were mixed, with gasoline inventories falling by 2.6m barrels – slightly more than expected, while distillate stocks unexpectedly rose by 3m barrels. Refinery utilisation increased by 1.5pp week-on-week to a robust 92.9%.

European gas prices remained volatile, with TTF briefly falling nearly 10% below EUR50/MWh yesterday before recovering most losses to settle 2.7% lower on the day. Prices remain elevated as Iran rejected a US-backed ceasefire proposal, keeping tensions high at a time when European gas storage levels are below 30% ahead of the refill season.

Middle East disruptions continue to ripple through energy markets, with gas prices still more than 60% higher since the conflict began due to the effective closure of the Strait of Hormuz and reduced LNG availability following damage at Qatar's Ras Laffan facility.

Saudi Aramco is reportedly set to supply around 40m barrels of crude to China in April, while deliveries to India are expected to total roughly 23m barrels, slightly below last month. The decline reflects disruptions in the Strait of Hormuz, forcing Aramco to reroute volumes via the Yanbu pipeline on the Red Sea, where export capacity of around 5m bbls/d remains well below pre conflict Gulf shipments. While the rerouting offers partial relief, it fails to fully offset lost capacity, pushing up costs for Asian importers and highlighting the conflict's growing economic impact.

Metals – Nickel jumps on Indonesia export tax plans

Nickel prices jumped on the LME by almost 3% on Wednesday after Indonesia agreed to introduce taxes on outbound shipments of the metal. President Prabowo Subianto approved tariffs on coal and nickel exports on Wednesday, with discussions still ongoing around the specific tax rates.

Indonesia accounts for more than half of global nickel production, meaning any policy shift has significant implications for global supply and pricing. The move comes as Jakarta faces mounting fiscal pressure from higher oil prices linked to the Iran war, with Indonesia a net importer of crude and refined fuels.

Indonesia has long pushed to move up the value chain by curbing raw material exports. For the global market, it reinforces supply side uncertainty and adds upside risk to nickel prices.

Agriculture - Sugar softens as lower oil prices weigh on ethanol demand

Sugar prices retreated by more than 2% from a five month high as easing geopolitical tensions

between the US and Iran pressured crude oil prices, reducing the incentive to divert sugar into ethanol production. With volatility persisting and strong output from key producers such as Brazil and India underscoring ample supply, the recent rally appears to have run ahead of fundamentals, limiting scope for further gains as oil linked support fades.

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Rates Spark: Aggressive central bank discount remains sticky

Markets appreciate the US seeking a deal, but without concrete steps, they will be reluctant to dial back their aggressive central bank discount. The ECB cannot retreat to just watching the data, but also must balance market expectations along the curve



Markets are pricing a more forceful ECB response, even as President Christine Lagarde emphasises caution and data dependence

Central banks balancing the market as tensions linger

Markets are appreciating US efforts to seek a deal, but without concrete steps, the overarching narrative of two to three ECB rate hikes this year remains unchanged. Even for April, market pricing still points to a 60% probability of a hike. The key variable to watch remains the oil price, and that hasn't really moved much since the start of this week. Headlines do bring some movement, but investors are reluctant to trade on every rumour related to a deal. And with Iran clearly feeling confident, the situation is still fragile and can easily escalate again.

Still, market pricing, especially for April, appears aggressive against the backdrop of President Christine Lagarde suggesting on Wednesday morning that the ECB would not act until it had enough information and could look through a short price shock, underscoring the differences to the 2022 situation. Of course, being still in the middle of the turmoil, one can only speculate about the duration of the disruption. A more forceful reaction by the ECB could still follow to nip any price

spiral in the bud. This is what the market is pricing – and it gels with the relative stability of longer-dated (forward) inflation swaps. But the market is also looking for a partial reversal as indicated by the hump, e.g. in the Euribor futures strip.

At this stage, the ECB is engaged in a balancing act of managing market expectations and the curve. If it sounds too doveish, the risk is that the long end runs away as inflation expectations become unmoored. If it sounds too hawkish, growth concerns take over. At some point, even meeting market expectations might be required to keep the balance – and buy time.

The strong decline in longer-dated gilt yields might provide an omen for EUR and USD curves. Increased worries about the UK growth outlook are pushing 10Y rates down faster than the short end. Markets are now positioned for at least two hikes this year, which would bring the policy rate to 4.25%, implying a very tight policy stance. In effect, the brake on the economy would eventually force a loosening of policy and could very well push the Bank Rate to below neutral over a longer horizon. If this becomes a more dominant narrative, also outside the UK, then we would expect a bull flattening of curves.

Thursday's events and market view

Markets remain driven by geopolitical headlines, and most data releases for now have been overtaken by events. In the eurozone, there will be some final Q4 GDP releases from the Netherlands and Spain. We will get the ECB's M3 data, while Germany will release a consumer confidence reading which could be more informative in assessing the current turmoil's impact. ECB speakers such as de Guindos and Muller, who are scheduled to speak, should also be more relevant than data for now.

In the US, we will get the usual weekly jobless claims data and manufacturing index from the Kansas Fed. The main focus here, too, will be the Fed speakers scheduled for the day: Cook, Miran, Jefferson and Barr.

From the Bank of England, Breeden, Taylor and Greene are scheduled to speak.

In primary markets, the UK tenders 30y gilts for £0.3bn, while the US Treasury will sell new 7y notes for US\$44bn.

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Monitoring Hungary: Tensions are rising

In our latest update, we share our thoughts on how the recent spike in geopolitical tensions could affect the Hungarian economy and the forthcoming elections. Rising energy prices pose a significant risk to the Hungarian macro landscape



Budapest, Hungary

Hungary: At a glance

- We downgraded the GDP outlook for 2026 from 1.9% to 1.7%, mainly due to energy price shocks and the expected worsening of consumer and business confidence. We see risks tilted clearly to the downside.
- February's retail sales and industrial data were positive, but the recent war in the Middle East could destroy all hope for a sustained acceleration in growth.
- Some companies have started laying off staff amid cost pressures from the labour market. Hence, employment dropped to a level not seen for almost five years.
- The external surplus is still holding up, but geopolitical risks are mounting due to higher energy imports and still limited external demand. We see the current account balance dipping into a small deficit in 2026.
- Inflation came in at a 10-year low rate in February, but the country can't enjoy this for long. The energy price pressures will lift inflation, and we revise our forecast from 2.6% to 3.4% for the yearly average in 2026.

- Following the first rate cut in 16 months, the easing cycle ended before it had even got started. The war in the Middle East has forced the National Bank of Hungary to revise its stance, pushing it towards the more hawkish side.
- February's budget deficit came as an unpleasant surprise, mainly due to a spending spree. However, revenues are also showing signs of being out of sync with economic data. We forecast a deficit-to-GDP ratio of around 5.5–6.0% for 2026.
- The general election is taking place on 12 April 2026. Some political analysts think that rising geopolitical tensions favour the ruling party, making the political race even tighter.
- With the central bank being as cautious as possible, and even actively using its FX reserves to redirect import-related FX market flows, if necessary, the EUR/HUF 385-390 range will act as a gravity line.

Quarterly forecasts

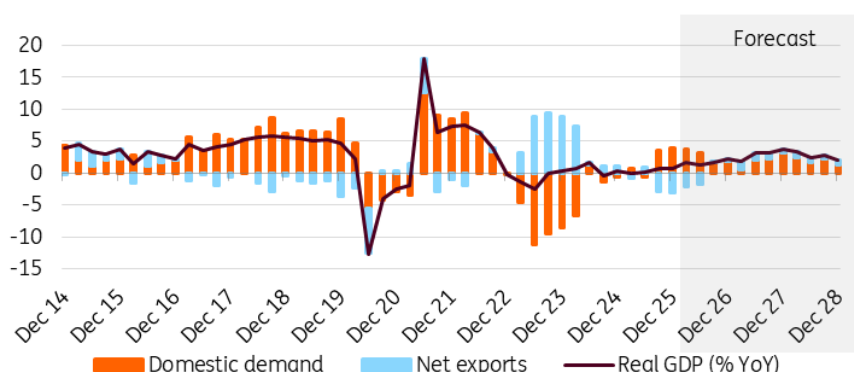
	4Q25	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F
Real GDP (%YoY)	0.8	1.6	1.4	1.7	2.2	1.9	3.2	3.1
CPI (eop, %YoY)	3.3	2.3	4.1	3.7	4.0	4.1	3.9	3.5
Central bank key rate (eop, %)	6.50	6.25	6.25	6.00	6.00	5.75	5.75	5.00
3m interest rate (eop, %)	6.47	6.45	6.25	5.90	6.00	5.75	5.60	4.85
10yr yield (eop, %)	6.77	7.35	7.05	6.35	6.70	6.60	6.25	5.50
EUR/HUF exchange rate (eop)	385.4	390.0	385.0	390.0	385.0	390.0	385.0	390.0
USD/HUF exchange rate (eop)	328.4	339.1	331.9	330.5	320.8	325.0	320.8	325.0

Source: ING

Geopolitical risks complicate the path out of stagnation

Although the detailed data for the Hungarian economy in the fourth quarter did not differ significantly from the initial figures, the outlook for economic growth in the near future appears uncertain. Spillover effects stemming from the structure of fourth-quarter growth and the energy price shock meant that we had to lower our economic forecasts for this year once again. Initially, we expected growth of 2.3%, which we later reduced to 1.9%. Yet it seems that even this was too optimistic. Currently, we forecast growth of 1.7% for 2026. This figure is subject to both positive and negative risks. On the one hand, consumption may be the driving force in the early part of this year. In the second half of 2026, we anticipate improved export activity, driven by new factories. However, the recent war in Iran tilts the balance of risks to the downside.

Real GDP (% YoY) and contributions (ppt)

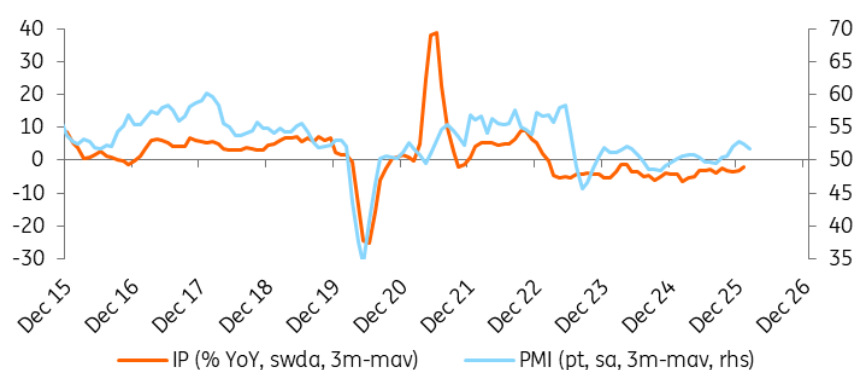


Source: HCSO, ING

Two good months in industry, but the outlook is cloudy

The performance of industry provided a pleasant surprise at the start of the year, by expanding in February. For the first time in a long while, Hungarian industry expanded for two consecutive months. Given the new export capacity that is becoming available this year, we can be optimistic about a lasting improvement. However, we believe this is more likely to occur in the second half of the year. That said, the ongoing war in the Middle East poses significant risks to industrial growth. As of now, we forecast that the fixed base index will recover and reach the baseline level – representing the 2021 average – by the end of the year. However, if the war drags on, we should expect the anticipated growth to be lower, and confidence indices could sink once again.

Industrial production (IP) and Purchasing Manager's Index (PMI)

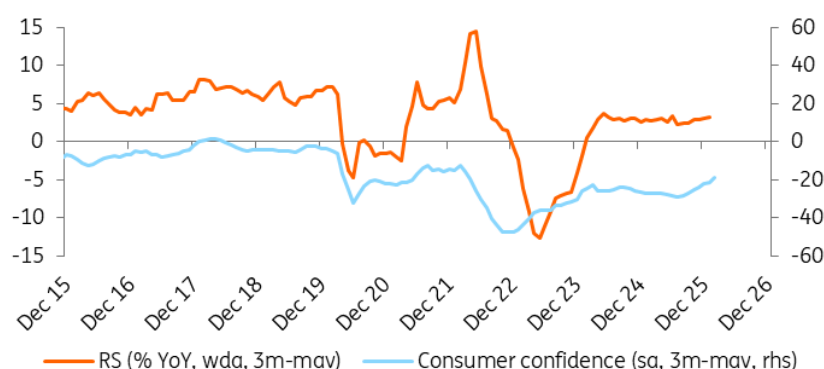


Source: HALPIM, HCSO, ING

Retail sales at risk of worsening consumer confidence

Despite the headwinds surrounding it, the retail sector started the year even more positively than forecasts had suggested. Domestic retail trade grew by 0.5% month-on-month in January, meaning that monthly growth has now been seen for six consecutive months, so performance is not faltering. While pre-election fiscal measures may have had a positive impact on spending levels, this is likely to be a short-term, one-off effect rather than a lasting shift in momentum. For retail sales growth to accelerate sustainably, a lasting improvement in consumer confidence is needed. Although this indicator has moved in a positive direction in recent months, the newly erupted war will be a significant setback. If consumption falls as a result, the only factor currently preventing the Hungarian economy from stagnating will be seriously jeopardised.

Retail sales (RS) and consumer confidence

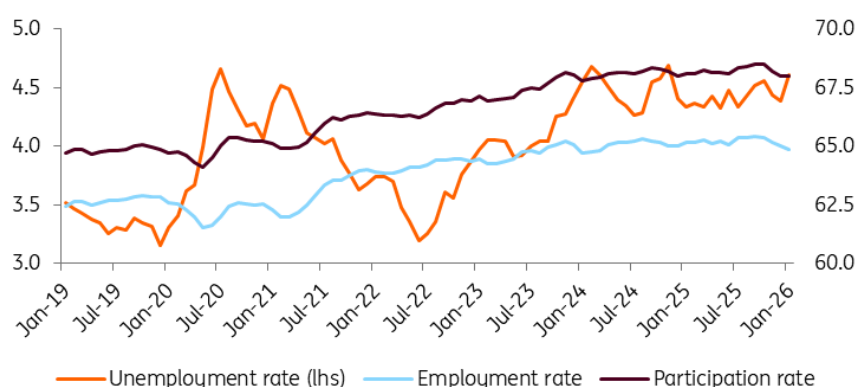


Source: Eurostat, HCSO, ING

Companies can't hold off any longer; layoffs begin

Unemployment figures saw a surprising increase at the start of 2026. It seems that companies adapted to the increased wage costs caused by the minimum wage hike much sooner than expected. According to model estimates, the unemployment rate reached 4.6% in January 2026. Data reveals that the decline in the working-age population continued at the start of this year, accompanied by a fall in employment. In January, there were just over 4.6 million people in employment, the lowest figure since May 2021. Since mid-2022, when the number of unemployed people was at a record low, the working-age population has fallen by 144,000. Due to supply-side issues, the labour market remains tight. External risks to GDP growth will put further pressure on companies, hence the rising probability of an increase in unemployment.

Historical trends in the Hungarian labour market (%)



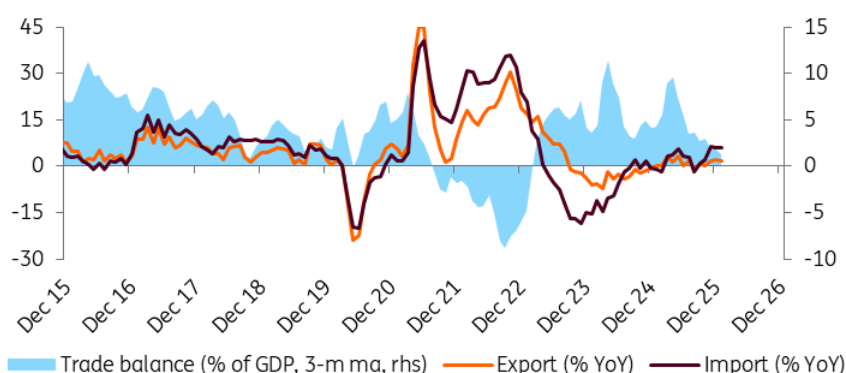
Source: HCSO, ING

Growth in imports, decline in exports: a bad combination

The trade surplus was EUR12m in January 2026, with the balance deteriorating by EUR756m compared to the same period in the previous year. The volume of exports was 9.9% lower and imports were 2.3% higher on a yearly basis. As large foreign direct investment (FDI) projects in manufacturing entered their final phase and energy needs rocketed due to the cold spell, imports increased. The lack of external demand kept exports in check. The outlook is somewhat mixed.

While some companies have decided to reduce the number of shifts due to retooling and demand issues, others have begun laying off workers. However, some new producers will gradually increase production and exports throughout the year. The war in the Middle East has negatively impacted overall confidence in the European Union, exacerbating the external demand issue. If the war drags on, we see the current account balance dipping into negative territory in 2026.

Trade balance (3-month moving average)

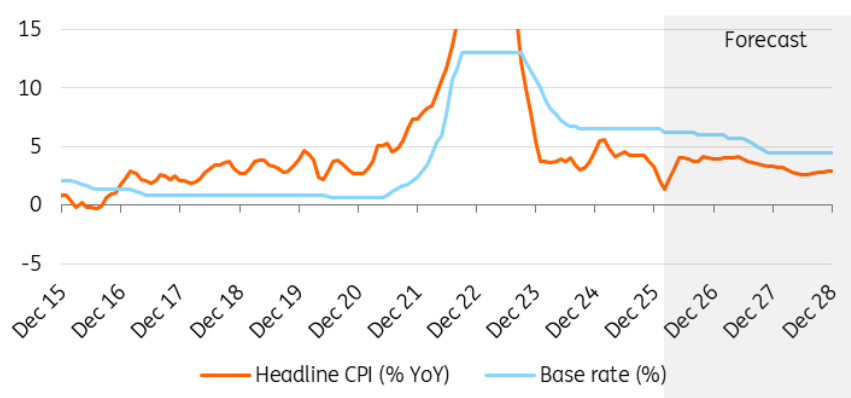


Source: HCSO, ING

Historically low inflation will prevent from extreme high levels

Inflation in February fell to a 10-year low. At 1.4%, the year-on-year inflation rate was lower than even the most optimistic forecasts. On a monthly basis, prices rose by just 0.1%. Food inflation was moderate on a monthly basis, while household energy prices fell significantly. Inflation in the services sector continued to develop favourably. According to our latest forecast, the year-on-year inflation rate could rise above 3% again by the spring and hover around 4% over the remainder of the year. Without the energy price shock, average annual inflation would comfortably have remained below 3%. However, given recent developments in the Middle East, a more realistic figure is now 3.4%, potentially rising to 3.8% in 2027.

Inflation and policy rate



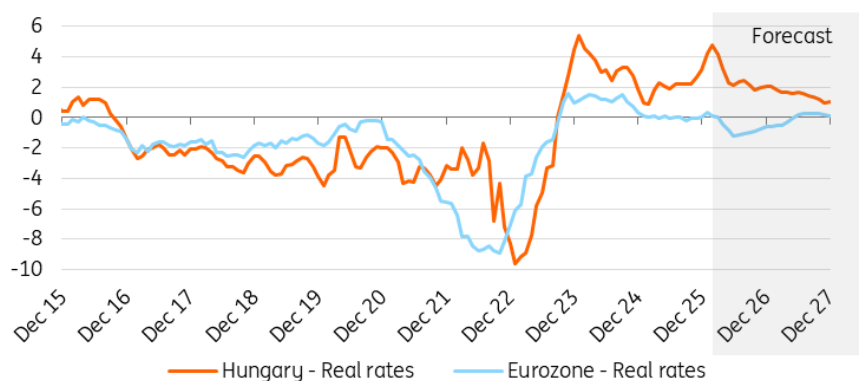
Source: HCSO, ING

Monetary policy is back to square one

In line with our expectations, the Monetary Council decided to keep the base rate at 6.25% on 24

March. Unsurprisingly, in light of the war in the Middle East and related market turmoil, the National Bank of Hungary has reverted to a hawkish stance. We agree with the central bank's view that it is too early to press the panic button. If our base case holds (40% chance) and the impact of energy prices fades as expected, inflation will remain mostly within the central bank's tolerance band. This could pave the way for monetary policy easing in the second half of 2026, following a lengthy pause. However, if our 'long war' scenario plays out (30% chance), we believe that the forint would require additional support, and the NBH could follow the European Central Bank's lead with the same number of rate hikes (most likely two) during the next couple of quarters.

Real rates (%)

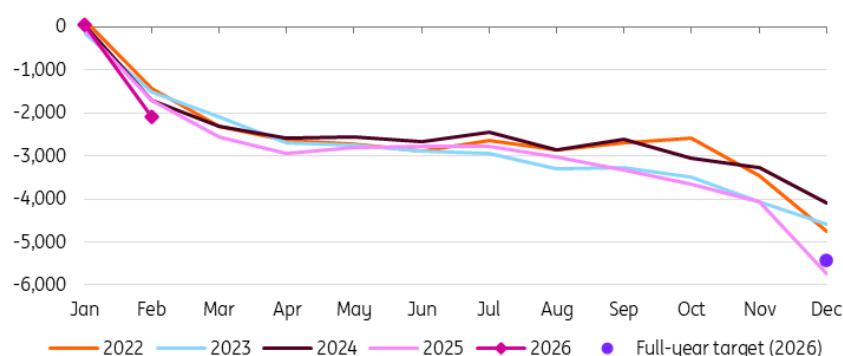


Source: ECB, NBH, ING

Election effect represented in budget data

By the end of February, the central government sector had recorded a deficit of HUF 2,107bn billion. This figure is the result of a small surplus in January and a record-high deficit in February. The high deficit in February was due to housing subsidies for public sector employees and wage increases for public sector workers, among other pre-announced spending measures. Fiscal measures aimed at boosting confidence ahead of the elections may continue to put pressure on the budget in the coming months. Based on the latest official data, the government is targeting a deficit of 5% of GDP in 2026. Taking into account the difference in the macro projections, we expect the deficit to be around 5.5% in the next couple of years. There is a significant risk of Hungary failing to secure the planned EU funds (including some RRF and SAFE money), which would put more pressure on debt financing.

Budget performance from cash-flow perspective (year-to-date, HUFbn)

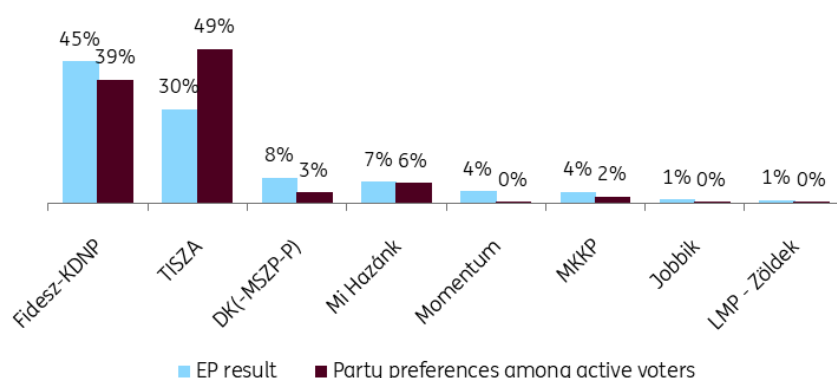


Source: Ministry for National Economy, ING

The political race is tightening

General elections will be held in Hungary on 12 April. According to most political analysts, the recent lead of the main opposition party (TISZA) suggests that it will be a close race, given the electoral system. Examining trends over the past month, the latest political analyses reveal a shift in campaign dynamics. Until now, the opposition party had dominated public discourse and steadily increased its lead, but geopolitical events have changed this. Escalating geopolitical tensions fit into the ruling party's (FIDESZ) narrative, enabling it to assert its messaging more effectively. These changes increase the likelihood of a quick narrowing of the gap in the opinion polls, creating a tail-risk scenario of a hung parliament or lengthy legal debates over the fate of certain seats, which could cause significant market volatility. Our macro forecasts are based on a no-policy-change scenario, in line with academic standards.

Average of opinion polls (early-Mar 2026) and the results of the 2024 EP election



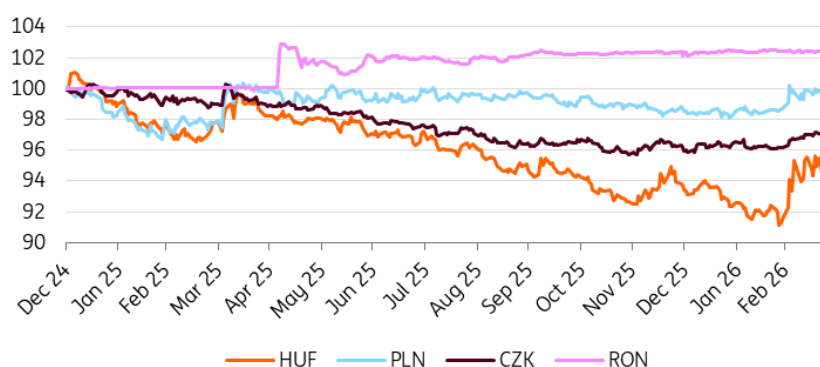
Source: EP, Opinion polls, ING

We foresee continued high volatility in HUF

We are now in a 'pick your poison' situation with regards to the forint. The war in the Middle East

and the energy price shock are hitting Hungary hard due to its reliance on energy imports. Even if the market calms down slightly, there is another red flag event: the general election on 12 April, which is set to be a close race. We think that monetary policy will be sidelined in the coming weeks, with either the energy situation or politics deciding the fate of the currency. With the central bank being as cautious as possible, and even actively using its FX reserves to redirect import-related FX market flows, if necessary, the 385-390 range will act as a gravity line.

CEE FX performance vs EUR (31 Dec 2024 = 100%)



Source: NBH, ING

On the political front, we predict that investors will react asymmetrically, as long positions related to political bets have largely been liquidated. We believe that there will be less resistance to a stronger HUF. In either case (a result-driven rally or a sell-off), however, the move will be short-lived, with the fundamentals dragging EUR/HUF back to around 385 as the dust settles on the election results.

Forecast summary

	4Q25	1Q26F	2Q26F	3Q26F	4Q26F	2025	2026F	2027F
Real GDP (%YoY)	0.8	1.6	1.4	1.7	2.2	0.4	1.7	3.0
CPI (eop, %YoY)*	3.3	2.3	4.1	3.7	4.0	4.4	3.4	3.8
Central bank key rate (eop, %)	6.50	6.25	6.25	6.00	6.00	6.50	6.00	4.50
3m interest rate (%)*	6.47	6.45	6.25	5.90	6.00	6.50	6.21	5.31
10yr yield (%)*	6.77	7.35	7.05	6.35	6.70	6.93	6.84	6.01
EUR/HUF exchange rate*	385.4	390.0	385.0	390.0	385.0	397.9	387.1	389.0
USD/HUF exchange rate*	328.4	339.1	331.9	330.5	320.8	353.2	330.2	324.2

*Quarterly data is eop, annual is avg. Source: National sources, ING estimates

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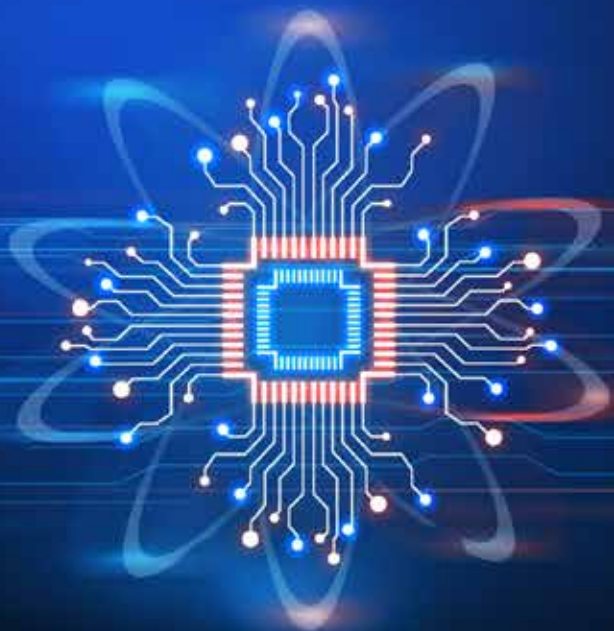
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MANUFACTURING INTELLIGENCE

Exploring the Spectrum of AI Use Cases

JUNE 2024



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The Increasing Appetite for AI in Manufacturing

Advances in artificial intelligence (AI) over the last 18 months have put AI on a new trajectory for manufacturing. Of course, the sector is no stranger to AI. Manufacturers have used machine learning for decades to teach equipment to perform tasks, and they are increasingly deploying deep learning to take on more complexity. Significant advancements in generative AI in 2022 opened an entirely new set of opportunities for using AI in the industrial domain. Generative AI is about far more than interacting with chatbots and rewriting essays. Building on deep learning architectures, generative AI can complement a manufacturer's machine learning and deep learning models to add even more intelligence to the entire value stream of manufacturing. All of the advances in AI over the last decade (computer vision, speech recognition, generative AI) combine

with other innovations in distributed computing, processing speed, edge computing, and open-source algorithms, to significantly advance manufacturers' journey toward Industry 4.0.

Most companies are early in the process of figuring out where and how to bring cutting-edge AI into their day-to-day operations. As Paul Bouvy, Senior Director of Supply Chain at **Owens Corning**, put it, "We have used AI every day for a long time to control our manufacturing processes. Here, I'm speaking about things like multivariable modeling and problem solving. But nowadays, everyone thinks of AI as just being about generative AI, large language models, and things like that. We're just starting work in that area."

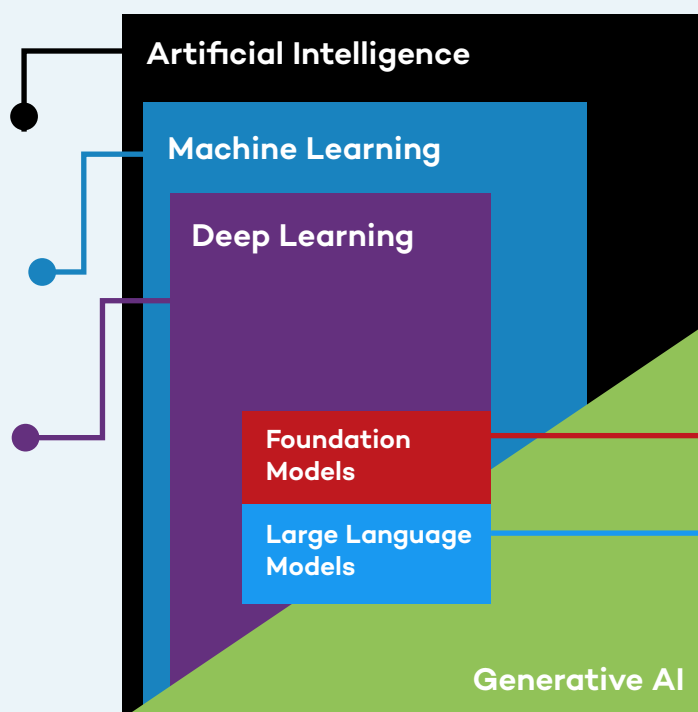
AI Definitions

Artificial Intelligence: Technology that enables computers and machines to simulate human intelligence and problem-solving capabilities.

Machine Learning: A form of AI that enables a system to learn from data rather than through explicit programming.

Deep Learning: A subset of machine learning that uses multi-layered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain.

Source: IBM: What's Next in AI?



Generative AI: Deep-learning models, such as chatbots, that can generate high-quality text, images, and other content based on the data they were trained on.

Foundation Models: Unlike task-specific models previously used by AI, foundation models are trained on a broad set of unlabeled data that can be used for different tasks. Early examples of foundation models are GPT-3, BERT, or DALL-E 2.

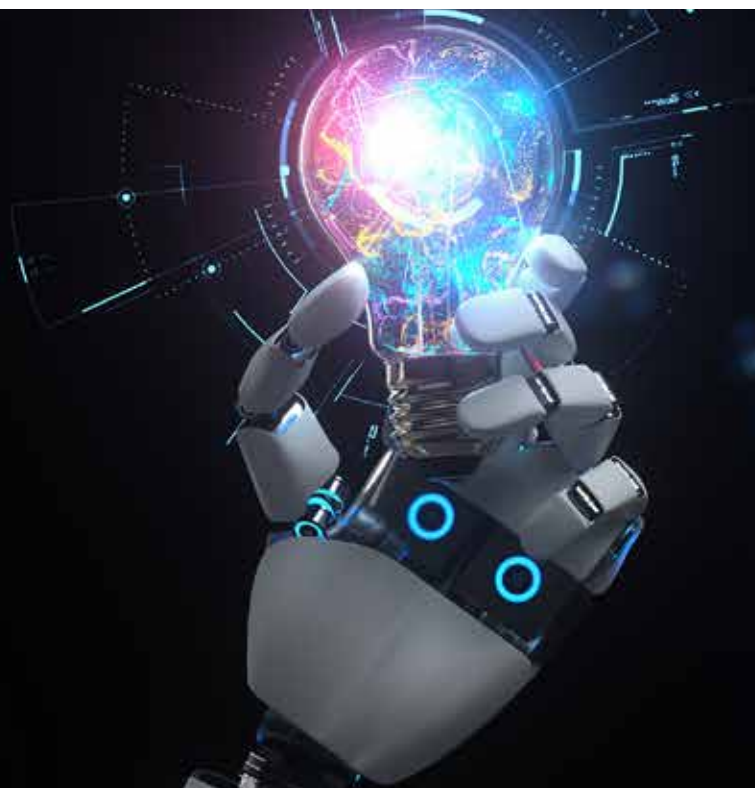
Large Language Models: Large language models are a category of foundation models trained on immense amounts of data making them capable of understanding and generating natural language and other types of content to perform a wide range of tasks.

What are the most important new use cases for AI in manufacturing and how quickly are manufacturers deploying them? Manufacturers Alliance Foundation took a deep dive into these questions to better understand the state of play for AI in manufacturing. We spoke with dozens of leaders across a range of manufacturing functions and conducted a survey of over 200 U.S.-based mid-cap to large-cap manufacturing companies.

KEY FINDINGS reveal a strong appetite for making progress with AI among manufacturing companies across dozens of new use cases already operational today. The vast majority (93%) have added new AI initiatives over the last 12 months, and another 6% plan to launch such initiatives soon. Significantly, many AI initiatives are geared toward strategic business gains, not simply cost or headcount reductions. Many manufacturers expect AI to deliver fairly rapid bottom-line improvements in productivity, throughput, and quality.

“We have just scratched the surface of advanced AI. We’re learning how to envision capabilities. Otherwise, we’re imposing limits on ourselves for no reason. If you can envision it with AI, you can achieve it.”

– Katrina Redmond, Executive Vice President and Chief Information Officer, Eaton

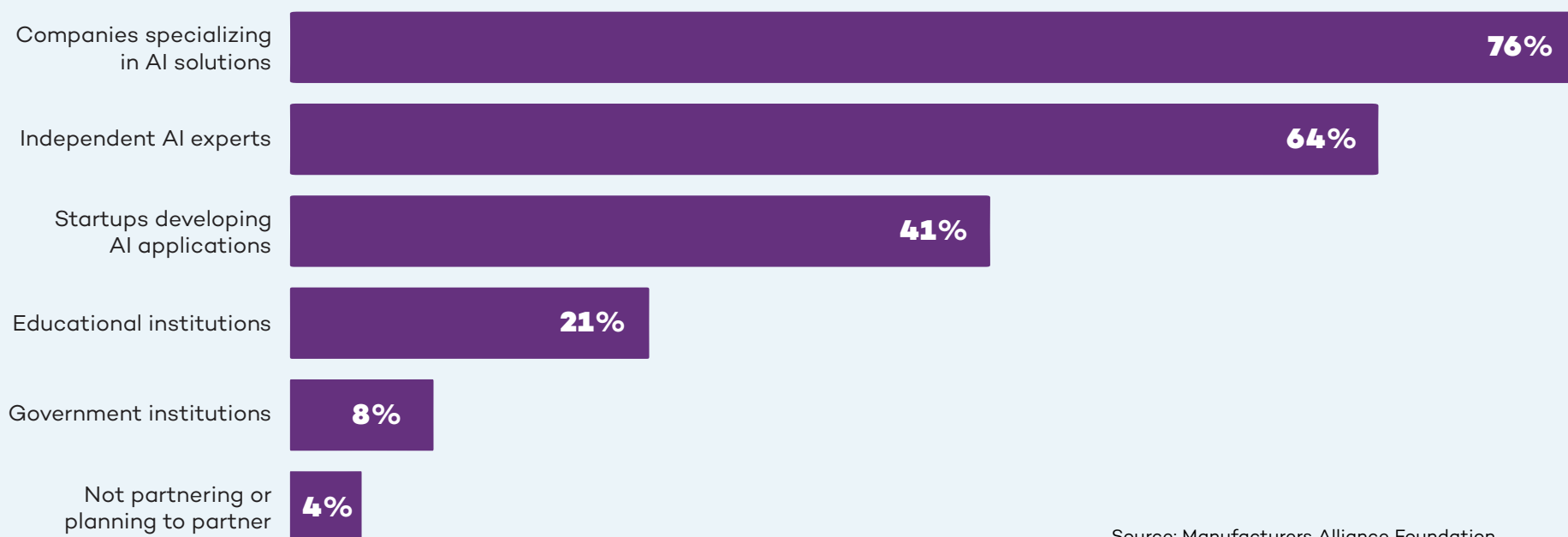


The vast majority (93%) have added new AI initiatives over the last 12 months, and another 6% plan to launch such initiatives soon.

The topline benefits, while expected to take more time, are the holy grail. That's why manufacturers have unleashed citizen developers (business users with little to no coding or IT experience) to come up with as many use cases as possible. As Wallas Wiggins, Vice President Global Supply Management and Logistics at **John Deere** told us, "One of the most exciting things is our citizen development effort where we essentially say, 'Hey, go figure out AI. Do something creative, even crazy. If it works, let's scale it across the entire business. We're trying to drill as many holes as possible because one of these days, one of them is going to pop oil."

The AI learning curve is steep for almost every organization. "I think one of the biggest near-term impediments is that you just don't think about using AI, even if you have it," John Bartho, Chief Information Officer at **Hyster-Yale**, told us. Most manufacturers are looking for guidance for the best use cases but more importantly, how to think strategically about AI and fit it into their technology roadmap. As Joanna Cooper, General Manager of the Mount Holly plant for **Daimler Truck North America** put it, "We can't make this technology transition with the attitude that we will do everything on our own. Daimler Truck has a vast number of partnerships with OEMs and other companies, including AI partners. We have to make progress together."

Where Manufacturers Are Partnering



Source: Manufacturers Alliance Foundation



AI for Topline Growth and Competitive Advantage

Finding new and more sophisticated use cases in the age of generative AI creates a new set of opportunities for manufacturers.

McKinsey estimates that generative AI can contribute between \$650 billion and \$1.1 trillion in annual productivity improvements for manufacturing and supply chain-related activities globally. With this much to gain, manufacturers are playing the long game when it comes to embracing new applications for AI. When we asked how they are prioritizing their AI initiatives, 55% said they are aligning these initiatives to strategic business objectives. Less than a quarter (24%) are focused on potential cost savings.

Manufacturers see cost as less of an obstacle when topline growth and competitive advantage are at stake. As Tim Speicher, Senior Manager of Advanced Analytics and AI at **MSA Safety**, told us: “The initial value is going to be in productivity, such as saving time. Then there are revenue driving opportunities where AI can help us – either in products or in services – and that’s where the excitement comes in.” When we asked manufacturers to rank how they plan to measure the ROI of

Top 6 Expectations for ROI

- | | |
|----|---|
| #1 | Revenue growth |
| #2 | Improved uptime |
| #3 | Product quality |
| #4 | Customer satisfaction |
| #5 | Long-term strategic impact |
| #6 | Performance relative to industry peers, competitors |

Source: Manufacturers Alliance Foundation

their AI implementations, revenue came in as the clear first choice. Long-term strategic impact and competitive advantage against peers ranked fifth and sixth. Cost savings trailed in ninth place.

Manufacturers are in it for the long haul. Very few (16%) expect a quick ROI in under 12 months. For the vast majority, the ROI time horizon is farther out: 42% expect to see it within one to two years, and 40% expect two to four years.

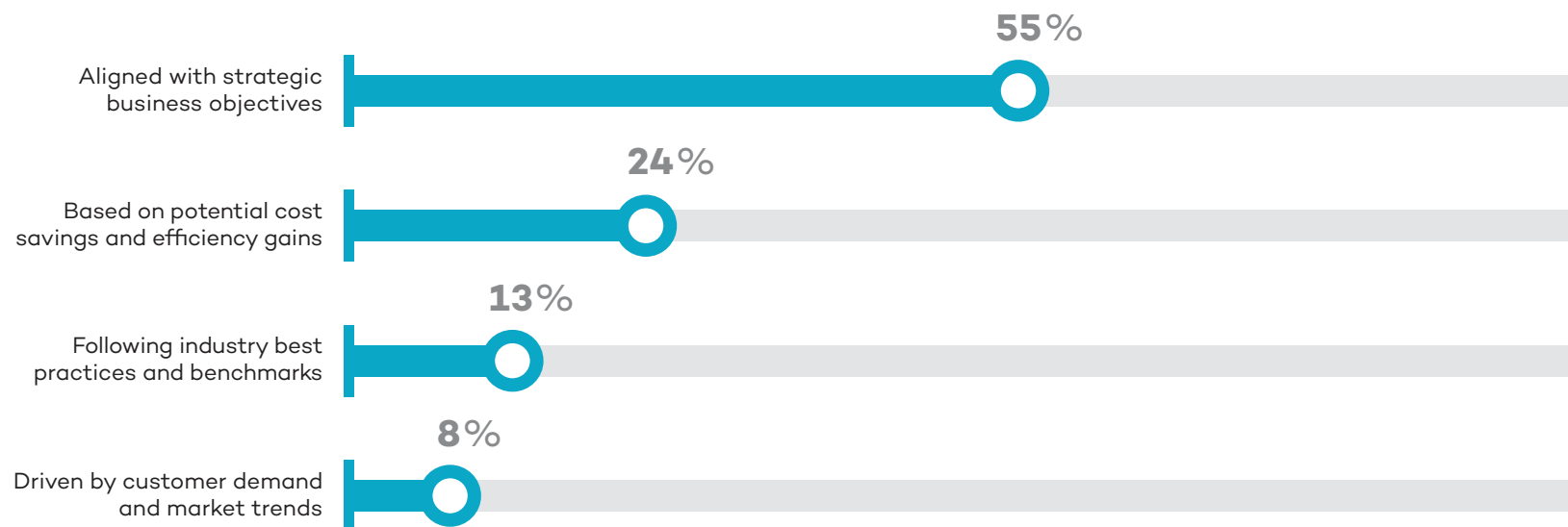
Given the magnitude of change, it will take time for organizations to absorb the impact from AI and realize the full benefits. Karen Powers, Digital Product Transformation Leader at John Deere, talked about just this aspect: “I think probably the biggest challenge in general is not determining where we want to use it, but how we build comfort with it. When will people be able to trust the results and allow the

system to take action for them? The change management aspect of all of this is huge.” Bringing people along will take some time. “We do have executive buy-in. Change is always difficult, so for the business to be totally behind it, we need some quick wins,” Tim Speicher of MSA Safety, told us.

There is a clear recognition that the latest advances in AI are real and here to stay. It will become table stakes just like previous technologies, such as e-mail, Wi-Fi, and texting. As Ben Grimes, U.S. Enterprise Commercial Vice President for Manufacturing at **Microsoft** predicts: “Once someone has it, they won’t want to give it up.” The record-breaking pace of consumer-grade adoption suggests this prediction may prove accurate. Already 27% of American adults say they use AI daily or several times per week, according to **research by Pew**.

AI Investments Focus on Strategic Gains

How do manufacturers prioritize AI implementation initiatives?



Source: Manufacturers Alliance Foundation

75+ Early Use Cases for Advanced AI in Manufacturing

Supply Chain Management	Design & Engineering	Production	Maintenance	Safety	Quality	Info Systems & Cybersecurity	Warehousing & Inventory	Aftermarket / Customer Services
Alternate component optimization	Buyer behavior prediction	Cobots	Anomaly detection	Employee fatigue and distraction	Assembly quality inspection	Attack detection systems	Automated picking	Aftermarket predictive maintenance
Faulty supply tracking	Controller engineering	Energy demand forecasting	Asset health monitoring	Employee PPE compliance	EHS standards library	Attack interruption systems	Autonomous vehicle delivery	Call center knowledge support
Inbound delay forecasting	Design coding assistance	Energy optimization	Contextual predictive maintenance	Potential safety risk prediction	Process parameter optimization	Automated IT operations	Demand forecasting	Connected product diagnostics
Intelligent supplier selection	Design for manufacturability	Machine controller project optimization	Maintenance planning	Real-time monitoring shop floor	Quality prediction	Cybersecurity automation	Dynamic container scheduling	Customer service chatbot
Raw material cost estimation	Design for quality	Material handling optimization	Predictive maintenance	Remote safety inspections	Quality standards library	Data cleansing	Dynamic load matching	Virtual assistance for tech support
Raw material optimization	Design for regulatory compliance	Production line balancing	Root cause analysis	Safety incident analytics	Regulatory compliance inspection	ERP system integration	Dynamic slotting	Warranty claims optimization
Supplier price comparison	Generative machinery design	Production line coding		Safety trend analytics	Visual inspection – cosmetic	Infrastructure performance tracking	Inventory optimization	
Supplier relationship management	Generative product design	Process mining for digital twin setup			Visual inspection – pass/fail	IT service issue prediction	Order processing chatbot	
	Instrumentation data capture	Production planning optimization				Master data management	Predictive stock replenishment	
	R&D efficiency	Robotic process automation				Risk detection optimization	Warehouse automation	
	Verification testing	Spare part inventory				Response automation playbook	Warehouse visibility	
		Virtual expert assistance						

Current AI Use Cases in Manufacturing

There are already hundreds of AI use cases relevant to manufacturing and many of them have passed the proof-of-concept phase to start delivering value. In the past year, **Siemens** has identified 300 generative AI use cases across all manufacturing domains. Del Costy, President and Managing Director, Siemens Digital Industries, US, told us about the company's cross-functional generative AI program which aims to identify efficiency gains, novel product features, and new offerings. "More than 70 of these are already under implementation in a Proof of Value," Costy shared.

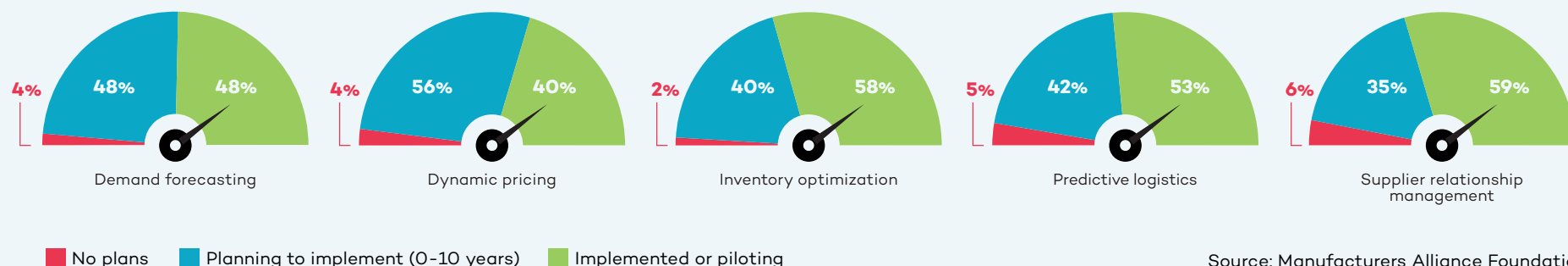
Manufacturers are moving quickly to add AI throughout the manufacturing value stream in areas such as:

- >> **Supply Chain Management**
- >> **Design and Engineering**
- >> **Production, Maintenance, and Safety**
- >> **Quality**
- >> **Information Systems and Cybersecurity**
- >> **Warehousing and Inventory**
- >> **Aftermarket and Customer Services**

This report focuses on seven areas, representing different points along the manufacturing value stream, where AI will grow in importance to manufacturing operations.



Supply Chain Management Use Cases



When it comes to trying new use cases for AI, most manufacturers view the supply chain as a good place to start. AI helps them collect data from across the supply chain in real time. Given the relatively recent disruptions in global supply chains, it is not surprising that supply chain management ranked as the number one functional area for AI initiatives currently among manufacturers we surveyed. “So much of what supply chain management does involves generating content, conversation, validation, and opportunity identification. So, machine learning and generative AI have a ton of opportunity here,” Karen Powers of John Deere told us.

Intelligent supplier selection is already being used to help evaluate a supplier’s pricing as well as resiliency based on current market data. Another application is **supplier relationship management**. As Paul Guerrier, Advanced Technology Center Manager at **Moog**, told us, “I can see us using generative AI tools to evaluate incoming information from vendors to check that the vendor data pack is complete, contains

good performance measurements, etc.” Tim Speicher of MSA Safety envisions AI helping procurement teams “negotiate better contracts and terms with our vendors.”

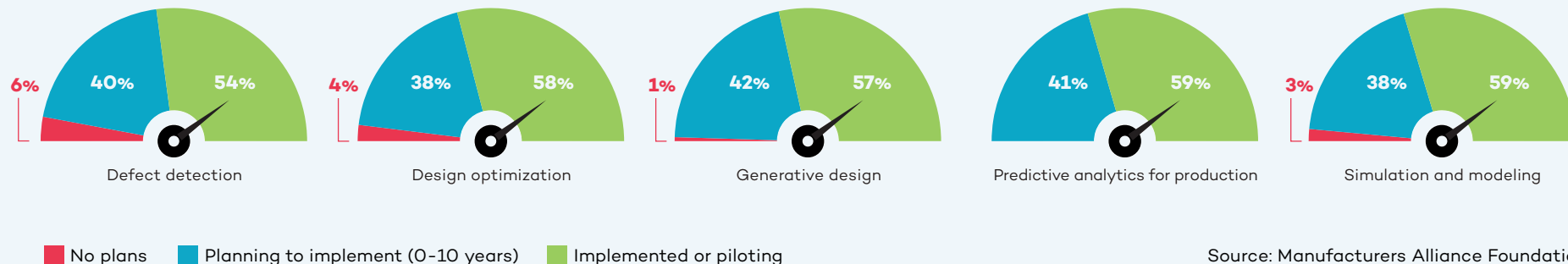
When there is a sudden global shortage of a specific material or subcomponent, AI can be used for **alternate material optimization**. Katrina Redmond, Executive Vice President and Chief Information Officer at **Eaton**, talked about partnering with **Palantir Technologies** to use generative AI for finding alternatives. “All of our information from our ERPs is processed by these large language models. A part shortage or a raw material shortage may be preventing us from completing the manufacture of a product. Do we have alternatives available? By having the descriptions in the system, you can sometimes match on description. If you need an 8-foot rod but only a 10-foot rod is available, we just have to cut 2 feet off to satisfy the requirement for the order.” Marc Rosen, Strategic Accounts, Palantir Technologies, added: “Working with Eaton, we created this AI agent that helps cross-reference parts so that if you have a

\$1 million sale being blocked by \$50 worth of raw materials, it helps them find what else they can use in their inventory to complete the order.”

AI for **raw material optimization** is helping steel manufacturers determine how to use their own scrap in the most efficient way. “As a steel manufacturer, we always have a mix of scrap material on hand as well as the possibility of what that scrap could turn into. So, we’re using AI to help us figure out the probability of producing the product that we want out of this mix. It will help us streamline both the ordering process on the raw material end and the manufacturing process on the melt end,” Jared Noble, Director of Digital Technology for **Charter Manufacturing**, told us.

Looking ahead, one manufacturer is hoping that advanced machine learning and AI will help them predict outcomes of a range of end-to-end supply chain decisions to help them land on the right choices based on the metrics they prioritize.

Design and Engineering Use Cases



For design and engineering, the current use cases are designed to speed up the innovation process and reduce time to market through things like AI assistance in **product design, innovation design, machine design, and regulatory compliance**. Michelle Drew Rodriguez, Partner at **Roland Berger**, discusses innovation efficiency in her work consulting manufacturing clients: “We have already identified more than 200 AI use cases for manufacturers, including in supply chain, procurement, operations, human resources, and more. For example, in the product development and engineering space, we have been working extensively

More than three quarters of manufacturers are starting to use AI for regulatory compliance.

Source: Manufacturers Alliance Foundation

with clients on optimizing product design as well as the product development process. We’re using AI to review the entire R&D portfolio, then systematically applying AI to advance R&D efficiency throughput and deliverables, including reducing the time to develop commercially viable products. This has enabled us to create efficiencies such as reduction in R&D spend (25%+) as well as greatly improved throughput (up to 60% reduction).”

One large manufacturer is using AI to look at **design for manufacturability** of a product to ensure that any errors in design are caught before the manufacturing of a new product or product version begins. Design for manufacturability is especially important in the consumer electronics manufacturing space, where miniaturization must be balanced against the angles and spaces that are required for assembly, especially if that assembly is being handled by a robot.

More than half (57%) of manufacturers are starting to use AI with generative design

and another 22% are planning to start in the next two years. Significantly, manufacturers with more than \$10 billion in annual revenue are much more likely to be focused on AI for generative design, R&D, and quality than manufacturers in lower revenue ranges.

Southco sees potential future applications of AI during the **advanced product quality planning (APQP)** phase. Deep learning AI algorithms can assist with designing for quality by helping the engineer select the right materials. Shailesh Patel, Director of Quality at Southco, described the situation: “If an engineer tries to design a product in a certain material, AI tells them ‘Don’t use that material.’ This is based on past data. We have had a ton of issues with some specific problem materials, so this is very important for us.”

Design for regulatory compliance is another promising application for AI. Southco has started exploring the potential for AI in this area. As Jim Ford, Vice President of Engineering at Southco, explained: “We have to sift through a lot of compliance

“We don’t have to spend hours writing the code. From a product development perspective, it’s huge. We use it every day.”

**– Tom Hower, Vice President of R&D,
Heavy Duty Transportation Group,
Marmon Holdings**

paperwork, and it is a very manual process with people parsing that data. So, we want to scrape the relevant information out of everything that is coming in so they can manage the compliance elements.” John Deere is also using AI to assist with regulatory compliance by providing AI-driven prompting systems that interrogate the regulatory documents to assist with the interpretation of standards.

Southco is just starting to look at the potential uses for AI in **generative product design**. “A lot of our products are very modular. So, we’d love a customer to come in and say, ‘Hey, I like this product, but increase that grip by 10 millimeters.’ And if we could do some of that through AI, potential savings and customer responsiveness for us would be pretty significant,” Jim Ford of Southco told us.



AI in Verification Testing

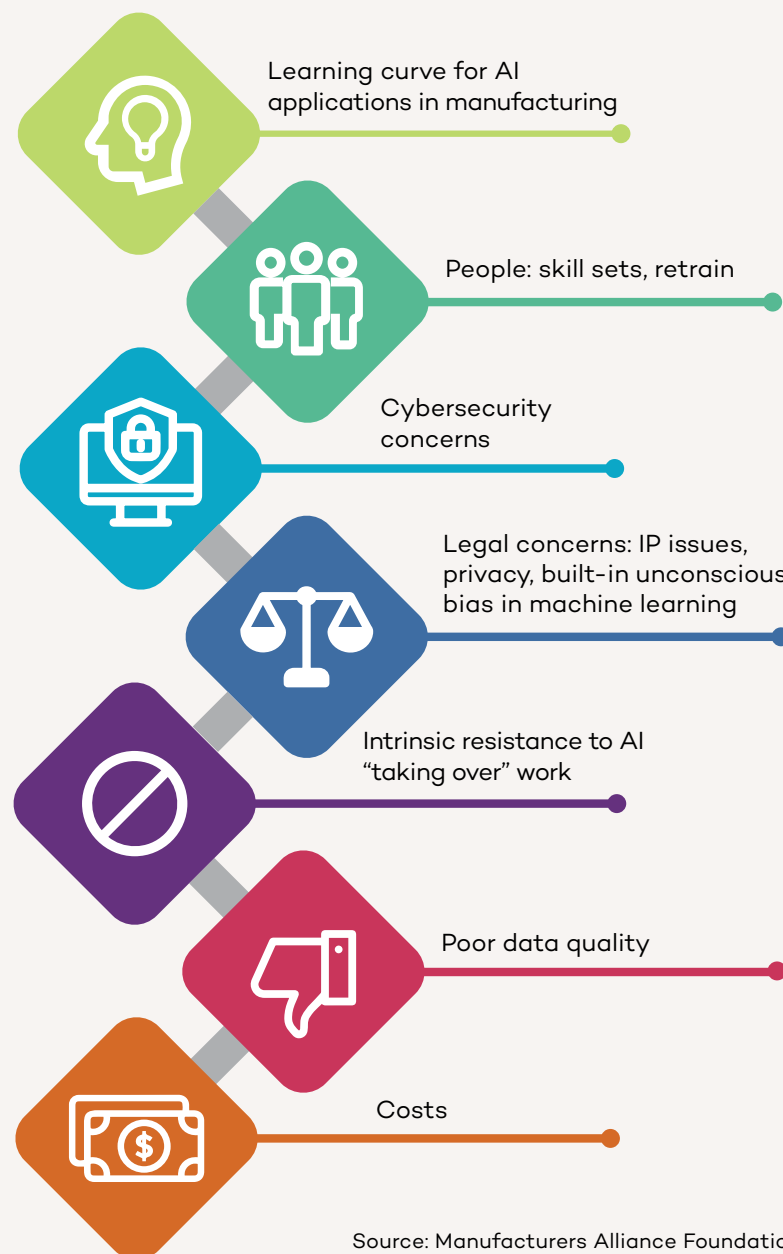
Verification testing is fertile ground for AI. More and more products have connectivity to other products, machines, or cloud applications, and this requires much more involved testing. The key is to develop efficient testing models. As Tom Salapow, Executive Director, Research & Engineering at MSA Safety, put it: “When you have devices connecting to the cloud and to each other, you have a bigger ecosystem and the instances of things you have to verify goes up exponentially. Instead of doing 10 tests, we now have to do 1,000 tests. So, we’re looking at how to develop design of experiments to narrow that down, so instead of doing 1,000 tests, maybe we can do 100 effective tests and cover everything. I think that’s where AI can help us.” Salapow pointed to the magnitude and speed of change in this particular area: “It’s something we’re evolving into now. We have built a team of R&D verification engineers. That wasn’t even a role five years ago. That’s one of the challenges they’re working on now. Any time you make a change, it has a significant effect across everything. You almost have to keep this thing running in real time as things are changing. And that’s a natural fit for AI.”

Several manufacturers talked about using generative AI for **product development coding**. As Tom Hewer, Vice President of R&D, Heavy Duty Transportation Group at **Marmion Holdings** explained: “One of the places we’re using AI is to help our engineers with coding. My mechanical guys are a lot better at software now that they’re using AI because they can say, I want a Python script that does this or that, and the AI generates a baseline script immediately that just needs to be adjusted. We don’t have to spend hours writing the code. From a product development perspective, it’s huge. We use it every day.” Ben Grimes of Microsoft has seen more than a 40% increase in code development among its customers using its GitHub Copilot, which helps developers generate code more quickly.

Looking toward the future, manufacturers are also thinking about AI not only for products but for **generative machinery design**. One manufacturer shared: “Can we use AI to help us design not just our products, but also the machines that make our products? Is AI going to come up with things that a human sitting down with a piece of paper is not going to consider, such as transformative, non-standard configurations of parts and machines?”

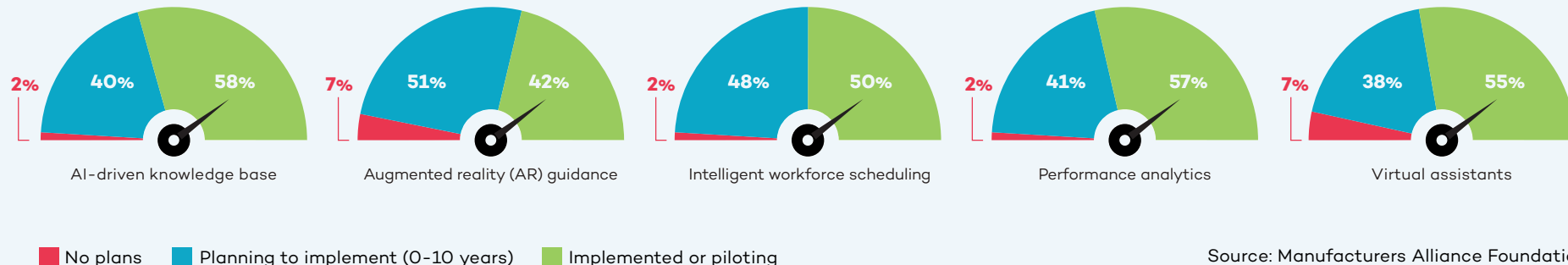


How Manufacturers Rank 7 Key Obstacles to Adopting AI



Source: Manufacturers Alliance Foundation

Production, Maintenance, and Safety Use Cases



The range of operations in the production and maintenance space opens up a wide variety of new AI use cases. Most manufacturers we spoke with have started using generative AI to provide **virtual expert assistance** to production and maintenance teams. Our survey also revealed that 55% of manufacturers are starting to use virtual expert assistants in this space currently and another 16% are planning to do so within the next two years.

Generative AI also has a role to play by providing detailed instructions for maintenance teams including lists of spare parts and images. This not only speeds up remediation but enables less experienced staff to address maintenance problems.

Clarios feeds a large quantity of engineering documents in different formats (PDF, PowerPoint, Excel) into an in-house generative AI system. Sami Khan, Senior Manager for IT Platform Services at Clarios, told us: “We wanted the AI system to ingest about 1,000 different documents rather than having people searching through hundreds

of thousands of pages to get an answer. The chatbot spits out a short answer based on the specific question that is asked. We also want this data to be refreshed periodically as people upload more documents so that it is never out-of-date.” Khan provided a specific example: “You can ask the system ‘what is the procedure for testing paste density?’ The system generates a short answer based on the documents that were ingested and links to the source of the answer. If you want to dive in further, you can click on the source.”

Krishnan Alampallam, Vice President of IT Applications at Clarios added: “We have the same equipment in multiple different versions all over the world, but we also have a constant churn of people. We never want to depend on an individual brain to deliver these answers because then it becomes tribal knowledge. How do we get rid of tribal knowledge? That’s the whole goal behind it.”

Coding assistance is another early application area for manufacturers. This is especially relevant for operations employees who have decades of hands-on experience

but little training in coding. Siemens and its motion technology customer **Schaeffler** are partnering to use generative AI to help Schaeffler’s **automation engineers generate code** for programmable logic controllers more efficiently. This enables engineering teams to generate code faster, with less effort, and with fewer errors. “In the past, we had to speak to machines in their language. With the Siemens Industrial Copilot, we can speak to machines in our language. In a few years, AI will be omnipresent in industry,” Siemens Digital Industries global CEO Cedrik Neike predicted.

AI-enabled predictive maintenance is raising the bar on how assets can be protected in an industrial environment. A great example is AI-based **asset health monitoring** for industrial robots. AI can monitor robot health as well as predict failures that might happen before a scheduled maintenance.

BMW is using AI in the assembly process to analyze data generated by its conveyor technology system. Without any additional sensors or hardware, the company transmits data to its own predictive maintenance

cloud platform where algorithms scan for irregularities that could bring production to a standstill, such as changes in power consumption, abnormal conveyor movements, or illegible barcodes. With AI, BMW is able to identify and avoid faults that would have caused 500 minutes of disruption per year at one assembly plant. With one vehicle rolling off the assembly line every 57 seconds, each minute counts. In addition, BMW is able to use the system's findings to continuously refine and improve the algorithms. The vision is for this system to learn to estimate the time between detection of a fault and potential stoppage, aiding operations teams in determining how to prioritize maintenance tasks.

When these types of artificial intelligence and machine learning are combined with generative AI, manufacturers are able to create a **conversational predictive maintenance** user interface facilitating a natural language interaction between the AI platform and maintenance experts. This dialogue streamlines the decision-making process and saves both time and money.

The synergy between AI and human capital is valuable. Del Costy of Siemens stressed the importance of integrating "human expertise and machine learning to extract relevant information from the equipment and process status." By analyzing the process around the equipment, it is possible to detect equipment behavior changes early, enabling better operating and maintenance decisions.

Given the power of AI in this space, it is not surprising that half of manufacturers are

starting to use AI for predictive maintenance analytics and another 25% expect to do so within the next two years. **USG** is using AI for "**anomaly detection** of both our assets and our processes," Lou Stocco, Senior Technology Manager at USG told us. "It identifies trends that are less than desirable. It could be an issue with mechanical, electrical, maintenance, or production. We also look at raw material consumption, energy usage, and all of the things associated with the overall manufacturing process. We're calling it condition-based manufacturing," Stocco said.

Advanced Technology Services (ATS) is already looking at what AI will be able to do in terms of **contextual predictive maintenance** as capabilities evolve. Micah Statler, Director of Industrial Technologies at ATS, talked about the next steps: "We're looking to go deeper into the predictive and the condition-based realm and feed our models more contextual data. It's one thing to say the machine is vibrating with this signature. It's a whole different thing to say it's happening under these conditions." Statler's vision is "to drive engineering change without the failure of the equipment to begin with. This prolongs the life of the asset."

In the industrial engineering phase, AI tools can help manufacturers and machine builders optimize their designed operating efficiency, desired throughput, and machine availability by running AI-based optimization tools on the simulated machine controllers (hardware in the loop simulation) in high-frequency. The German company **plus10** provides **machine controller project optimization** software helping companies shorten ramp-up

phases significantly by deriving throughput potentials on a controller step-chain level at a very early phase. This reduces later-on mean time to repair (MTTR), production variability, and overall machine ramp-up time of complex production or assembly lines. CEO Felix Georg Mueller told us, "Our software is learning live on machine controllers, either simulated ones during design phase or later on real-ones integrated in the line. By learning behavior models based on high-frequency data streamed in milliseconds, we formalize the behavior, reason the models, and provide counter actions in case of stops, disturbances, short stops, or scrap parts. That generates plus 10% or more output on average, hence our name." When one machine is behaving differently than another in terms of output or frequency of failure, an additional optimizer product can determine the root cause. "We can benchmark machines with each other even if they are thousands of kilometers apart to learn behavior and optimize on the go. In the past, this had to be done manually by engineering, improvement teams, or system integrators," Mueller said.

Manufacturers are also evolving from using AI with individual machines or lines to large-scale **production planning optimization**. Eaton's Katrina Redmond talked about the challenges and the solution. "We have 30,000+ different products. We always need to know how, when, and in what bundles they are selling. AI has given us better clarity across all of our different ERP environments to know which items we are clear to build. For example, it tells us if we have all the product available in the quantities necessary to fill order X, Y, or Z. So, AI is really helping

the planners and the production floor teams figure out which products they can complete before getting to step one of the manufacturing or assembly process. Essentially, AI is telling us, 'Don't bother with this order yet because you're going to be missing a piece at step four.' That allows us to go to the next item where we have all parts and material available."

Material handling optimization is another promising area for AI because it brings together a wide range of data points such as geography, mobility, wireless communication, supply chain, and more. More than half (57%) of manufacturers we surveyed are starting to use AI for material handling and 22% have it on their radar to implement within the next two years. Hyster-Yale is using AI for kitting. John Bartho described what are essentially "shopping cart kitting areas where people are building kits for assemblies which are scheduled to come into production next." The impact is significant in terms of throughput, quality, and cost. "We identify parts shortages earlier and maximize the efficiency of the assembly line. The quality improves because the less you interrupt the process, the higher the quality," Bartho added. This is an entirely new operating model for Hyster-Yale: "Increasingly, our suppliers are either onsite or across the street. This has a financial implication because we're not paying until consumption. The historical model is to buy stuff, put it on the shelf (where it becomes working capital), and then consume it. By using AI for process modeling and relying on more nearby materials, we're more productive and have more working capital efficiency," Bartho said.

Safety plays a role in every function, but it is particularly critical in the production and maintenance domain. AI-based visual systems are already on the market that address everything from **personal protective equipment compliance** to employee fatigue. **Stroma**, for instance, provides edge cameras and industrial edge mini-computers for advanced vision-sensing capabilities to ensure operational staff are wearing the appropriate PPE. Stroma's customers in the energy sector, for example, monitor the use of face shields, gloves, and footwear in real time. "We are tracking safety procedures to ensure they are properly followed in the field. In addition, our system can warn field operators in real time and send those same warning messages to managers," Anil Uzengi, Co-Founder and CEO of Stroma, explained. Stroma's systems can also check for **employee fatigue and distraction**. Uzengi shared an example from a customer in the consumer packaged goods sector: "You can see the worker's attention level is 82%. If it drops below 70%, the system issues a voice alarm to the worker in real time with a customized warning like 'Go grab a coffee!' The worker can respond via the same device to confirm the message was understood. Warnings are captured on a dashboard to measure event trends over time."

One Environmental, Health and Safety (EHS) Director talked about the value of AI-powered **remote safety inspections** in locations where an EHS manager is not physically present on site. "We run extremely lean as a company and we're a smaller manufacturer in terms of operations, so we just don't have the resources to have EHS at every single plant.

AI can help us supplement that, for example by identifying a hazard in an aisle, a blocked fire extinguisher, or inappropriate PPE. I think it's fantastic." They also talked about using AI for **safety trend analysis**. "With AI you can discern from the data if you have a location issue, a people management issue, or a supervisor who is not performing."

Mitigating potentially catastrophic safety events is a priority for every manufacturing company. At **The Heico Companies**, Vice President of Environment, Health and Safety Dave Roberts is focused on **safety incident analytics** with respect to potentially severe incident or fatality or PSIFs, a commonly used measurement in the EHS space. "We're trying to prevent catastrophic events from happening, and we're using AI to help us identify the critical precursors that lead to those major events. The whole idea of tracking PSIFs is to avoid getting bogged down by chasing minor events and be in a better position to prevent a significant incident," Roberts said.

Future safety applications based on AI are an important research area for MSA Safety. As Tim Speicher pointed out: "PPE is really the last line of defense. The first line of defense is not to let something bad happen to begin with. And in between there's all this space where we could use more of the data that we collect from our connected devices. We may be able to predict or project bad behavior before it starts. I mean, we're not talking about the movie *Minority Report* here, but we can start to identify problem areas, problem tasks, or even problem people. That's where our opportunities are."

AI as a Training Accelerator

Most of the companies we interviewed are struggling to keep up with training requirements. Generative AI is being used to address training challenges including new employee onboarding, cross-training of existing staff, and bridging the knowledge gap between early and late career employees.

As Joanna Cooper of Daimler Truck North America pointed out, “AI is an awesome tool, especially in light of the changing workforce. There aren’t a lot of people that come from a mechanical background, people aren’t working on their cars much anymore, and shop classes don’t exist in many schools right now, although some schools are starting to offer them again. Being able to use AI to help employees understand more quickly what good looks like as quickly as possible really helps.”

Felix Georg Mueller at plus10 talked about the ability of AI to reduce the reliance on tribal knowledge when production line issues occur:

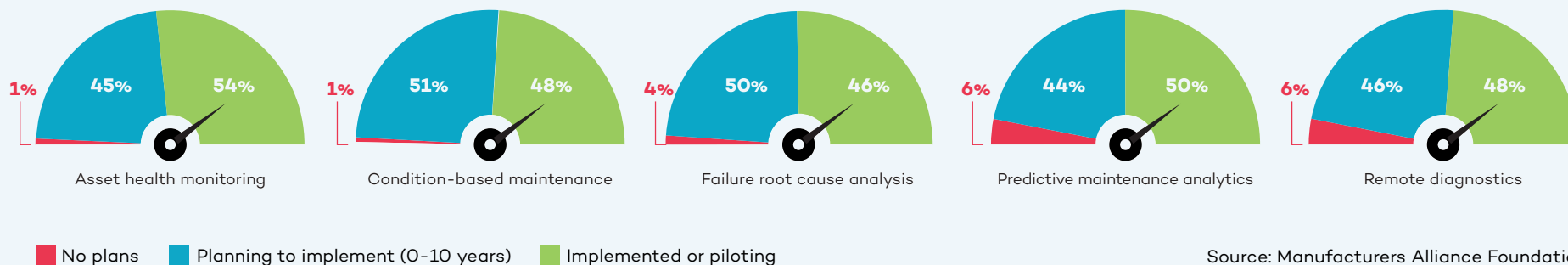
“Let’s say you have an expert who is sitting in front of the line who knows everything by heart. Then you have no problem because he or she can solve the problem in minutes. But those people are quite rare, especially on night and weekend shifts. Employees may not know where to find the root cause or how to address it. There can be a lot of trial and error, and sometimes they make the situation even worse. So our software tells them the root cause related action that needs to be taken and reduces the mean time to repair.” As a result, less training is required because plus10’s optimization tool proactively provides the missing knowledge, which reduces problem-solving time up to 40% in real-world, multi-shift operations.

A leader who manages training for a product design engineering team in the off-highway vehicle industry described the situation at his company. “More than half of our design engineers have been with the company less than five years. Three to four of those years have been spent working from home. So, we’re looking at ways to help these new engineers find the information they need to be successful. In our engineering knowledge world, AI can go out and scour whatever we connect it to and bring that information back. The idea is to teach these R&D engineers how to do their jobs based on our standard processes and tools, but without having to go through years’ worth of development with somebody sitting right next to them.”

In addition to AI-based training, manufacturers are also deploying “train the trainer” campaigns to spread AI knowledge throughout their organizations. Jared Noble at Charter Manufacturing told us about training “30 AI subject matter experts inside functional areas across the company’s business units.” Those people are helping to identify valid use cases for AI including the value it can deliver and the shape of a deployment. “The goal is for these people to have a good understanding of the basics of the tool sets as they’re seeing problems in their space. For us in the data science group, it means we’re getting more valuable information about how to approach use cases. We want to avoid having a divide between the technologists and the people in the function,” Noble stressed.



Quality Use Cases



AI is adding significantly to the technology mix supporting quality. The advances in computer vision over the past decade have already made computers superior to humans in their ability to catch things that are not visible to the human eye. Now, with generative AI, analysis of images can help reduce false positives (good products that fail inspection) as well as false negatives (bad products that pass inspection) thus increasing throughput and quality. More than half (55%) of manufacturers are starting

More than half of manufacturers are starting to use AI for defect detection systems. Another 20% plan to within the next two years.

Source: Manufacturers Alliance Foundation

to use AI for defect detection systems in production. Another 20% plan to within the next two years.

Daimler Truck North America is using AI to ensure **vehicle assembly quality**. “Engineering designs are becoming more complex, so AI enables us to ensure proper torque, orientation, and seating when parts of the truck are being joined,” according to Joanna Cooper of Daimler truck. All of these things need to come together in a timely manner on the line, so it is important that operators understand the desired result. “We’ve taken one of our more complex joints that has the layers of all the things that need to happen correctly to show operators what good looks like. We started with the vision system, and now we’re using machine learning to capture various failure modes to make the vision system even more robust. The idea is to capture those defects as quickly as possible,” Cooper continued.

Chatbots play an important role in quality management as well. “I’m frankly stunned at how good the chatbot technology is. We

John Bartho of Hyster-Yale talked about using AI for **visual inspection and analysis**: “I recently visited our factory in Mexico where we make all of our frames. You might think this wouldn’t be the most automated place, and historically it hasn’t been. There are about 300 employees, 210 of them are welders. But what I saw there surprised me. There are robots doing welding because they can do a seam without interruption thus delivering a higher quality weld. In the quality area, they are using vision systems to do random inspections of frames. The vision system compares the manufactured frame to the CAD drawing and highlights anything which might be variant. This is incredibly helpful because problems are often not visible to the human eye or difficult for the non-trained person to see. As we do more of this, we can improve quality overall by identifying these issues earlier.” As this example makes clear, the AI-assisted visual inspection and analysis is addressing not only quality but also a training bottleneck, an issue encountered by many of the companies we interviewed.



put all of our **quality and EHS standards** and processes into the AI system over a period of a couple of weeks. Now we have an open library of easily accessible information,” Martin Smith, Vice President of Quality and EHS at **Danfoss Power Solutions** told us. He also shared: “People are falling off their chairs at how simple it is to find answers. We tested it in a live meeting by throwing questions at it. It just came straight out with the correct answer. There was silence in the room. This saves literally hours of searching and sometimes only finding irrelevant documents.”

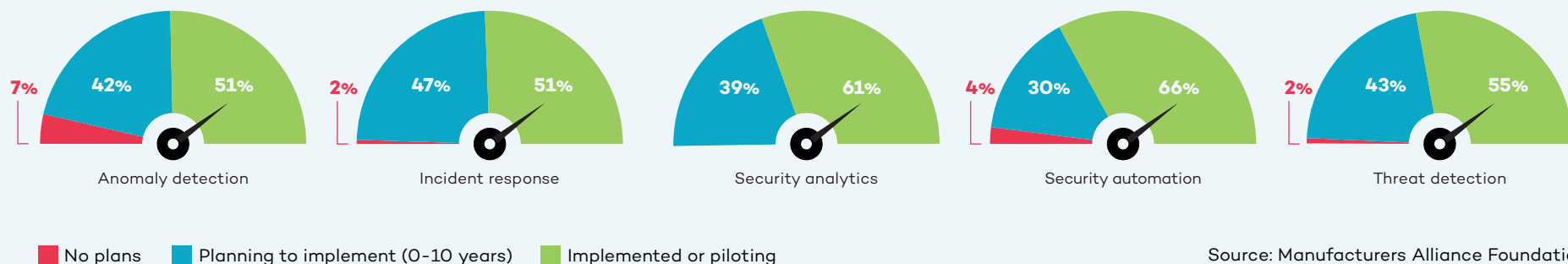
Computer vision for **quality prediction** is another valuable AI use case. Anil Uzengi of Stroma discussed his company’s work for a major tire manufacturer: “We are analyzing rubber sheets and detecting defects in real time. Our advanced vision capabilities use industrial edge cameras, and AI predicts anomalous areas. This triggers a warning to an operator or manager about the need to shut down the machinery.” Uzengi highlighted the incremental but significant ROI impact delivered by AI in this application: “This

company already enjoys 96% to 98% accuracy in their defect detection process, so we are just adding another 0.5% to 2% into this 96% to 98% accuracy. The incremental improvement in quality provided by AI may sound small, but for a large company like this, it means millions of dollars per year.”

AI can also take the subjectivity out of some visual inspection ratings, such as **cosmetic inspections**. For example, a specification for a bright zinc coating can be tricky without AI because humans may disagree about the definition of bright. Similarly, pass/fail cosmetic inspections of machine components can also be a matter of interpretation. Paul Guerrier of Moog shared: “In aerospace many parts are anodized, and pass/fail inspections look at things like dents and scratches.” It is virtually impossible to prevent them completely, but they can be minimized. Guerrier talked about potentially using AI “to define the exact criteria (size of flaws, for example), for pass/fail decisions. I think you can actually define that within the vision system.”

Moog started using AI for quality to increase throughput and to address cross-training requirements for its operators. Paul Guerrier described layering computer vision and deep learning into traditional visual inspection practices: “I would describe it as **AI-enabled inspection**. Our operators have traditionally checked things through a microscope. We added a camera as well as a computer to do the deep learning. We then created the hardware data set and the training images which were used to train the AI. Now the operator can either look down the microscope or they can look at the flat screen, which is displaying the image from the microscope in pretty much real time. If they look at the flat screen, the AI gives them pass/fail advice while they’re inspecting. The human is still totally in charge.” As Guerrier described it, this example of using AI is his “Goldilocks scenario” because it is an augmentation, not a replacement. “If the AI stopped working or started misbehaving, the operator still has control and we can always revert to manual,” Guerrier said. Moog has seen a significant productivity improvement because of this change. In addition, ergonomic safety is enhanced because employees can spend less time bending to look through a microscope.

Information Systems and Cybersecurity Use Cases



Digitalization is increasingly turning manufacturing companies into data-driven enterprises. Security and data quality are paramount and growing more challenging as manufacturers add more sensors, edge devices, autonomous vehicles, and other connected equipment to the factory floor. AI can help manufacturers address this complexity as well as monetize their data.

The key first task for most of the companies we interviewed is addressing the widespread problem of data integrity. **Data cleansing** is job one because data must be accurate, complete, and in context to be useful. As John Bartho of Hyster-Yale put it: “We have a lot of data, but when we first put it all together in the same pot, the stew didn’t taste good. Now we have a centralized group for master data management. We started with all the data around the customer, and we’re expanding that.”

Vitaalic’s Director of AI, Bob Straight, talked about aligning programs to fit an organization’s data strategy maturity: “I work

closely with our data team to make sure that I’m not overcommitting us if the data isn’t ready.” AI is helping companies address their data quality and data strategy weaknesses. AI-driven **master data management** can automate the monumental tasks of profiling, cleansing, and standardization.

“We are only scratching the surface in what generative AI will mean to security operations.”

– Kevin Faulkner, Director of Product Marketing, Fortinet

This is a critical step because the data will be used to train the AI model. If data is inaccurate or mislabeled, the output will suffer. One manufacturer shared, “We

created something like a **customer service chatbot** tool that uses AI to generate answers. Basically, the whole system worked great, but a lot of the answers were wrong because our source data wasn’t in the right place, was inaccurate, or wasn’t up to date. The weak link in our tool was the data that it was pulling from.”

Many companies have grown through acquisition and need to make disparate information systems work together seamlessly. Katrina Redmond shared Eaton’s application of AI to address **ERP system integration**: “We have so many feeder plants with different systems, and the information doesn’t clearly flow together. But with AI you can load all the individual ERPs into a system and then cross-reference everything more simply.”

Getting systems to connect while securing them against cyberattack is essential. Two-thirds of manufacturers are using AI-based **cybersecurity automation** to proactively defend against attacks before there is a breach.



AI and machine learning are well-entrenched in this space and have been “for more than a decade in the form of processing billions if not trillions of data points every day,” Richard Springer, Director of Marketing for OT Solutions at **Fortinet**, told us. “The conversation around generative AI is relatively new, but that’s the future. It’s not going to replace humans in this space, but rather make them more powerful and efficient.”

Fortinet has been adding generative AI assistants to its products since late 2023. Kevin Faulkner, Director of Product Marketing at Fortinet, explained, “this is not about opening a second browser window for Chat GPT and asking questions. This is actually embedded within our products. It is context relevant and context aware, including very

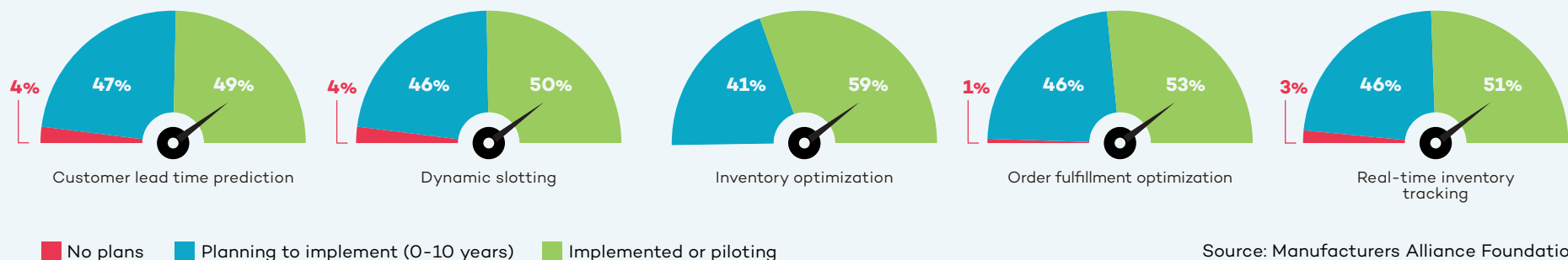
specific OT knowledge.” In the case of an **intrusion detection**, an employee would typically perform **threat analysis** including determining how dangerous it is, what it typically does, and what techniques can be employed to investigate the intrusion. “With our solutions, employees can use either a written or verbal interface to get answers to these questions. This makes people better at their jobs by allowing entry-level analysts to get more work done at a more senior level because they have an assistant looking over their shoulder,” Faulkner continued.

Large language model integration also makes it possible to generate recommendations on how to prepare and respond. As Rod Locke, Director of Product Management at Fortinet, told us, “You can actually build **automation playbooks** based

on best practices for a given scenario, including processing certain types of events in an automated way in the future.” The result is effectively upskilling employees in terms of their effectiveness to the organization.

Time is critical to security. When bad actors find out about vulnerabilities, they target them very quickly. “There used to be about an eight or nine day time period to prepare after a vulnerability was released. Now, that has been shortened to about four or five days,” Richard Springer told us. “It can be a very manual process to separate the wheat from the chaff and find out what’s actually going on. With generative AI tools, we’re making employees faster and elevating them to a higher-level response allowing us to shrink the time it takes to get in front of bad actors.”

Warehousing and Inventory Use Cases



Advanced AI is enabling manufacturers to solve a number of warehousing, inventory, and logistics challenges in ways that were previously impossible. **Warehouse management automation** systems are changing the way inventory is tracked, picked, and replenished.

A great example is **inventory optimization**. More than half (59%) of manufacturers are starting to use AI for inventory optimization and another 26% plan to within the next two

years. When a manufacturer announces a coming price increase, the result can be a spike in orders followed quickly by a backlog. To avoid this situation, manufacturers are using AI to prevent this type of avoidable inventory volatility. Bob Straight of Victaulic told us that inventory forecasting is a focus area for his team: “Addressing inventory forecasting challenges represents a significant win for Victaulic because it benefits not only operations and supply chain, but also sales and customer care.”

USG considers inventory management a strategic component of its value proposition as a company. **Dynamic slotting** plays a key role. As Lou Stocco explained it: “We are a make-to-stock manufacturer, so our guiding strategy is to be the lowest landed cost producer in all of our product spaces as well as the easiest to do business with. We are using AI to better align the amount we produce with the limited space in our warehouses. The idea is to get as much product to as many customers as possible



“We are transitioning these people from non-value-added work to value-added work which allows them to focus on parts of the operation that really need it.”

– Lou Stocco, Senior Technology Manager, USG



while giving them excellent service. For some of our customers, we guarantee next-day delivery. That level of service requires careful orchestration of a lot of product on the shop floor.” The big picture for USG is to have employees engaged in more valuable activity instead of extracting data, creating dashboards, or doing analysis. Stocco explained: “We are transitioning these people from non-value-added work to value-added work which allows them to focus on parts of the operation that really need it.”

When it comes to logistics, AI is playing new roles as well. John Deere, for example, is looking at AI solutions to become more efficient with its delivery trucks. Using generative AI for **dynamic load matching** may be the answer. As Wallas Wiggins of John Deere explained, “We would like to have a truck out there to deliver and pick up multiple times before it circles back home. I see that as one of the next big fronts for us because we spend a lot of money on logistics. We also have sustainability goals. I want to make sure every time a truck rolls, it’s meaningful rolling, not empty rolling.”

Process mining is playing a key role here. The German software company **Celonis** has developed process mining software that creates a digital twin of a company’s processes by capturing data from every split second of a process and turning it into a digital twin. Their work with **Mars** on dynamic load matching has achieved significant results. Mars has been **able to combine truck loads**, cut shipping costs, and speed up delivery resulting in an 80% reduction in manual touches as well as lower costs and reduced emissions.

Aftermarket and Customer Services Use Cases

Many manufacturers are exploring how to build deeper relationships with customers through service excellence and as-a-service offerings. AI can play an important role by giving manufacturers insights through **connected product diagnostics** and **always-on self-service assistants**. These tools give manufacturers actionable data about their products (how they are performing, how customers are using them) that can be plowed back into R&D and product development.

Traditional technical support is getting faster and better through **virtual expert assistants**. Siemens has deployed technical service chatbots that are able to understand questions and simulate human conversation. Katrina Redmond of Eaton talked about the challenge of technical support for products that are 40 or 50 years old and still in use. “We’re using AI to quickly ingest all of these manuals and decades worth of information so the tech team has it at their fingertips. Because of AI, a technical support person can answer a phone call from a customer and immediately respond with the best answer for their particular issue. Turnaround times can be dramatically reduced.”

One heavy equipment manufacturer is using AI for **warranty claims**. An executive there shared: “There’s a lot of free-form text in the warranty claims, several hundred words in each claim. Individuals used to spend an enormous amount of time reviewing these claims. Now we have deployed a large language model that simplifies the claim

information. We have just rolled this out, but it has the potential to save hundreds if not thousands of hours of people’s time looking through warranty claims per year.”

Danfoss is testing AI to omni-channel customer feedback arriving via email, call centers, and customer portals. “In the past, addressing customer feedback meant manually searching for product number, serial number, manufacturing site, point of sale, etc. I thought deploying AI to handle this would be hard, but it was remarkably easy. We have the same amount of information as we had before and didn’t lose anything,” according to Martin Smith at Danfoss Power Solutions. This allows employees in these roles to be re-deployed into value-adding parts of the organization.

Aftermarket predictive maintenance is another promising area for manufacturers. New entrants such as Tesla and Rivian are offering the ability to predict when a part might fail based on data they capture from the vehicle itself. This information can be enriched based on actual miles driven, vibrations, geography, climate, road conditions, and driver behavior. By correcting problems earlier, customers will likely save on repair costs and automotive companies will encounter fewer warranty claims.

Astec Industries is putting more and more intelligence into its products along what it calls the rock-to-road value chain. Eric Baker, Vice President of Astec Digital, talked about the importance of speeding innovation

and making its products smarter “because customers are demanding it.” One example is inspection of asphalt silos being used by customers involved in building or maintaining roads. “There is a lot of weight in those silos, and the material is abrasive. Silos need to be inspected to ensure that they maintain their structural integrity,” Baker said. With a robot or drone remote control device, Astec’s customers are able to visually inspect silos and process the information using AI. “Previously, analyzing that information had been a manual process, but with machine learning and AI, we are able to identify thin spots or worn-out welds to determine if a silo is structurally sound.” Baker talked about the growing importance of digital twins for this traditionally low-tech space. “We are investing heavily in augmented reality so that it is possible to see one piece of a plant that exists in the physical world and add other pieces in the digital space, allowing customers to visualize how things fit together,” Baker said.

Caterpillar is using aftermarket predictive maintenance to connect dealers with customers’ equipment. Its AI systems are analyzing **data from more than 20 different sources**, such as equipment sensors or fluid analyzers, to create an accurate picture of the performance and health of the customer’s asset. This allows a Caterpillar dealer to monitor the equipment, detect possible issues, and alert the customer to take action before a failure occurs. The dealer can then recommend the parts and services needed to keep the equipment running.

Getting Started and Creating an AI Roadmap

A Conversation with Michelle Drew Rodriguez, Partner, Roland Berger

Q **Manufacturers want to know how to deploy AI as quickly as possible. What do you recommend when you consult with manufacturing clients eager to get started?**

A First, we try to make the process as easy and pragmatic as possible – “less hype, more results” is our motto when it comes to implementing AI with manufacturers. We take them through a systematic framework for developing an AI roadmap, and partner with them to 1) assess their current state, 2) define their future state, and 3) create a detailed execution plan. (See roadmap, page 28.) At kickoff, we ask key questions including: What is the current state of their strategy, value proposition, and technological maturity? How are the people, processes, and culture working together in the organization now, and what does its wider ecosystem look like? We ask that companies be brutally honest in their answers; it is the only way to make progress. At this point, we often hear about problems with legacy systems, data silos, and poor data quality in general. This is very typical of many manufacturing organizations right now.

The current state assessment is a great way to get everyone on the same page and using the same vocabulary, which is an important first step. We see barriers to adoption for the simple reason that people may not have a clear understanding of what AI is, or what it can do for them. There’s so much hype around AI right now, and to many people it just seems big and nebulous. At the same time, we encourage companies to start small and find success quickly. What we mean is, once you’ve identified a few easy, low-hanging use cases within the organization, execute pilot projects to gain momentum. Quick success and real proof points help make AI tangible for employees. This is the quickest way to demystify AI and spark interest among people who don’t have much knowledge or belief in it. Demystifying AI and building a strong, unified quest for change needs to happen

early on. This allows companies to build leadership resolve on a strong foundation, including a commitment to invest and drive accountability through the businesses for the changes ahead.

Q **The current state exercise sounds important but also arduous. Do you find that companies want to stop here and clean up the existing problems, or do they want to plow ahead with AI?**

A From our experience, it’s a mix of both. Companies are finding that AI isn’t so much a new challenge but more of a solution to many of the problems they have identified in their current state analysis. Take the siloed data problem. AI can help manufacturers share data from disparate systems across multiple entities to make more sense of, for example, supply and demand trends. They get better at fulfilling customer needs and optimizing inventory instead of guessing or simply creating more buffer.

When companies start to envision their future state with AI playing a role, they also develop a clearer AI vision aligned with broader business goals. For example, they are not just brainstorming where they can add AI because it is technically possible; rather, they are prioritizing use cases based on business impact. This makes it easier, at this point in the process, for manufacturers to derive technical requirements and blueprints as well as needed organizational and governance structures.

The final step is execution planning. This is where manufacturers have an overall actionable plan as well as methods and KPIs to track the value that AI is delivering. They are making further refinements in terms of technology, people, processes, and culture, and are also documenting early wins and weaving any lessons learned back into the overall plan. Once they start gaining traction, companies can take a hub and spoke approach to saturate the businesses and functions with the knowledge and success already

gained. We find that creating and celebrating the early, quick wins really helps the broader organization's receptivity to implementing additional AI measures.

Q Based on your experience with manufacturers so far, how fast will AI spread in manufacturing?

A Nobody has a crystal ball, but I think it will be very similar to other technological transformations and will evolve at a much faster, exponential pace given the VUCA (Volatile, Uncertain, Complex, Ambiguous) times manufacturers live in now. Simply put, the fundamental need for real-time decision making is greater than ever before, and the stakes perhaps higher than ever. Therefore, I believe the typical barriers to adoption are lower for AI than prior technology adoptions. Use cases that make sense now will continue to make sense, but the solutions keep getting better and more all-encompassing. For example, use cases today are often aimed at solving one problem, but as AI maturity advances, manufacturers will be increasingly using platform-based solutions. They'll address multiple use cases at once rather than singular pain points. The impact will be enormous not only in terms of competitiveness of individual companies but also in the ability of manufacturing as a whole to take on megatrends such as resiliency, resource scarcity, and climate change. Manufacturers I speak with care about these issues and view digitalization and AI as key to addressing these grand challenges.

AI Roadmap for Manufacturers

	Current State Assessment	Future State Identification	Execution Planning
Strategy & Value Proposition	Secure leadership support & commitment Evaluate underlying aspirations from AI integration Understand business needs & expectations Gauge investment commitment to drive AI initiatives	Define and communicate clear AI vision & strategy aligning with broader business goals Establish evaluation and prioritization metrics for AI use cases, including applicability and business impact Assess ROI of AI initiatives	Formulate overarching actionable plan for AI integration (PoC, prioritized waves, etc.) Maintain, evaluate & prioritize the living document of AI use cases Develop methods & KPIs to track value delivered
Tech & Execution	Conduct readiness assessment of existing AI technology & processes > Technical blueprint re: architectures > Data availability, quality & governance > Infrastructure & security > AI tools & processes	Define target technical blueprint based on clear guiding principles Identify potential deficiencies/gaps as well as necessary tech partner(s) Establish data assets required for AI tools Design target infrastructure & forecast compute demand with aimed security level	Develop a roadmap to address the deficiencies within the technical blueprint Establish data acquisition processes Determine protocols to address data security & compliance risks Design a scalable infrastructure strategy to support future growth
People, Processes & Culture	Review as-is operating model Identify roles dependencies Examine present governance & policies Measure workforce capabilities and skills Evaluate established partnerships	Develop target operating model, including business process, roles, etc. Design AI governance structures Determine new skills due to AI integration Establish prioritization criteria for potential partnership	Establish tracking KPIs and milestones Integrate guidelines and principles into AI deployment strategy Design talent attraction & upskilling programs Activate prioritized partnership for implementation

Source: Roland Berger



Conclusion

As our interviews made clear, the race to bring more AI into manufacturing is on. It is not a question of if manufacturers will lean more heavily on AI throughout their value stream but how much and how quickly. The already large number of AI use cases is expanding every day as manufacturers open the aperture on how to think about the power of AI throughout the value chain. Likewise, as AI evolves from individual use cases to bundles of use cases that address sets of requirements, the power of AI will expand even more rapidly.

While there is a learning curve for every company, most organizations discover early on that AI helps them solve existing problems (such as training bottlenecks and supply chain issues) more efficiently while increasing safety, productivity, and quality in their operations.

At first glance, AI may seem like another major transformation to address within organizations that are already struggling with digitalization. But **many of the foundational steps required for successful digital transformation are also required to deploy advanced AI.** A mature data strategy leveraging accurate data will be the first major step for many companies and is a precondition for both digitalization and AI. Relatedly, many manufacturers find they need to take a hard look at their existing processes early on so that they avoid the trap of simply replicating a bad set of processes in the

digital realm. Connectivity is also paramount. Artificial intelligence, like digitalization, requires that machines and systems talk to each other in real time to provide full transparency into operations.

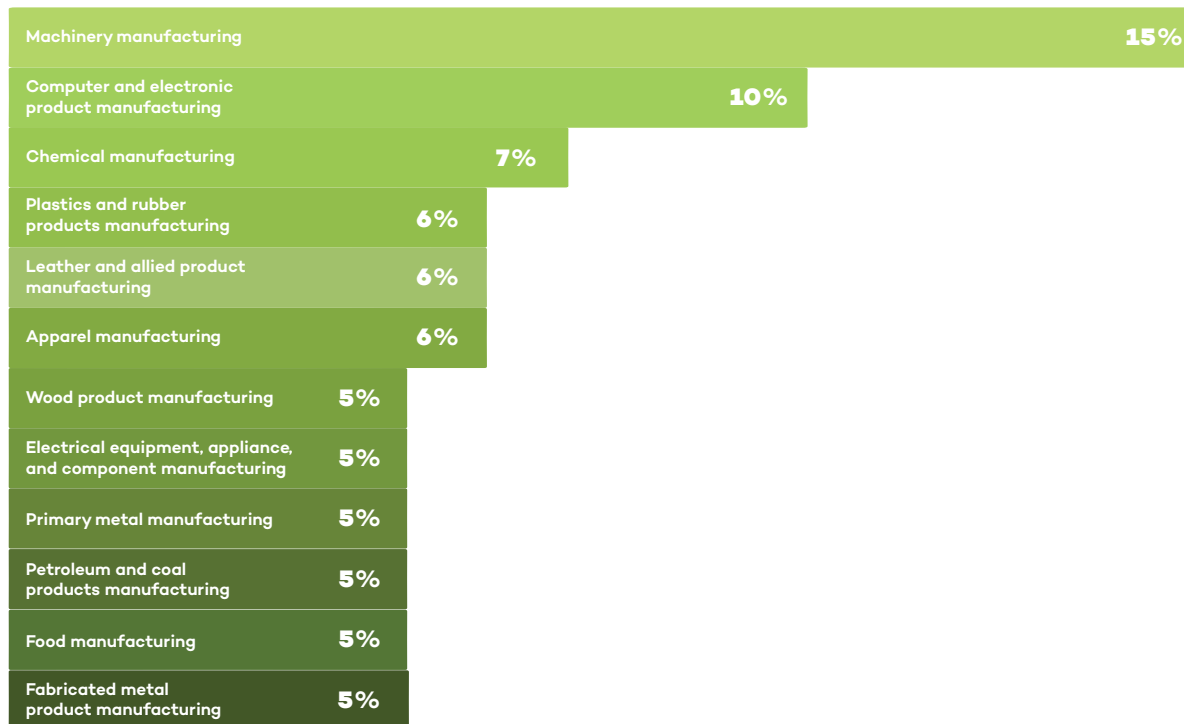
As with any technological transformation, the role of human acceptance is critical. Every company we spoke with stressed that AI represents a means to make their employees more productive and offer them more fulfilling tasks. It is critical for leaders to recognize the opportunities that AI brings and communicate these opportunities to employees. “Part of the leader’s role is to drive a confidence level into the team and say, ‘this can happen, it will happen, and there will be a culture change,’” Martin Smith of Danfoss Power Solutions stressed.

As Joanna Cooper of Daimler Truck North America put it, “it’s all about upskilling people to add value in a much different way. It’s a fallacy to say that AI won’t eliminate some jobs. It will. But it won’t necessarily eliminate the person’s ability to add value to the organization, and those are two very different things.” It is incumbent upon every executive to stress this point to employees. “It’s important for leaders to communicate the role of AI so that people understand it will impact their world, make their world better, and shift what they will be able to do,” Cooper added.

About the Research

Manufacturers Alliance surveyed more than 200 leaders in manufacturing to better understand the state of AI in manufacturing currently and in the near future. We have highlighted some statistics about the respondents.

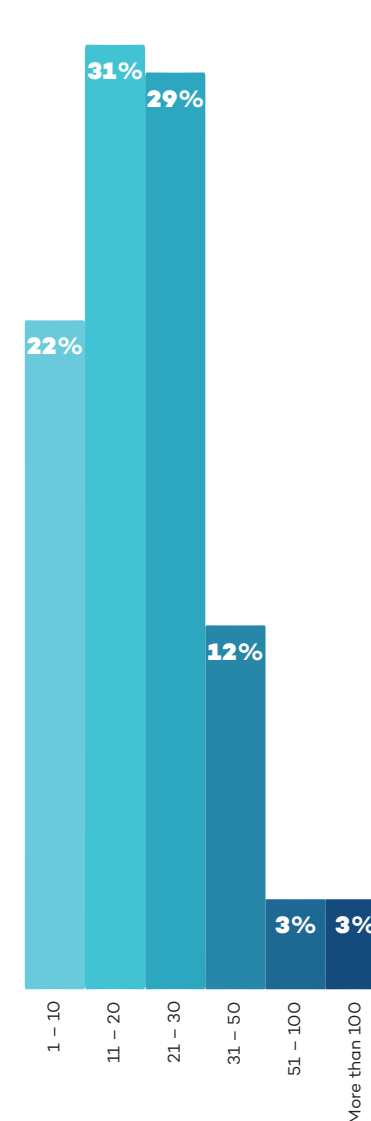
Top Primary Industries



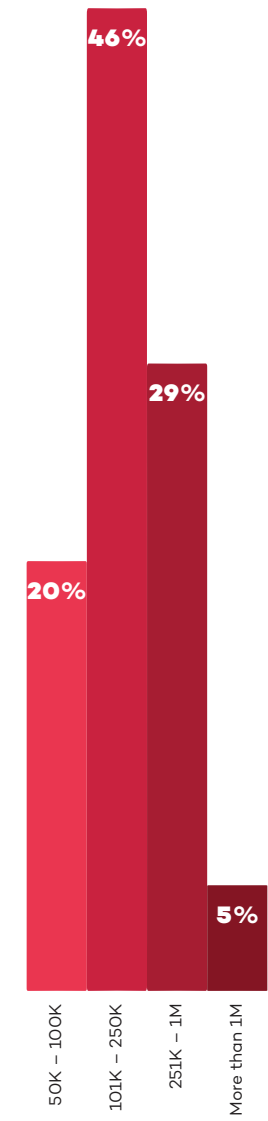
Annual Company Revenue



Number of Factories



Average Factory Size (in Square Footage)



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Watch: Why we still think EUR/USD can reach 1.20

One of the biggest questions we get from clients is how we can be so constructive about the euro, despite what's going on with Iran, the US, and the subsequent oil and gas shocks. ING's Francesco Pesole explains our FX team's thinking.

Read more in our latest FX Talking [here](#)



Why we still think EUR/USD can reach 1.20

We still think, despite everything, that the euro can rise to 1.20 against the dollar this year. Our FX Strategist, Francesco Pesole, says our team believes the Fed will look through this inflationary 'bump' driven by rising oil and gas prices and will cut rates twice again this year. And that will have an amplified effect on EUR/USD for several reasons. And while oil prices are set to remain elevated, we don't see the same risks for gas in Europe. So, the euro may trade lower in the short term, but should recover fairly strongly.

[Watch video](#)

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ING's updated forecasts amid energy disruption

Our economists and strategists have updated their forecasts to reflect the rapidly changing events in the Middle East and our new base case scenario



Oil tankers and cargo ships are being forced to reroute as the Strait of Hormuz is effectively blocked

Interim forecast update

The war in the Middle East continues, and the Strait of Hormuz is practically blocked. Our initial base case assumptions from our [March Monthly](#) regarding the war have become outdated. We are normally not in the business of permanently changing our forecasts, but given that our next Monthly Update is only due in April, and we think that these times currently require some guidance, we have updated our full set of [forecasts](#).

Three new scenarios

We have developed [three new scenarios](#) to guide our thinking and macro and market forecasts. In summary:

- **Scenario 1** (our new base case): Intensive combat will end within the next two weeks, but will be followed by lower intensity strikes that linger for several months. It's a scenario in which the opening of the Strait of Hormuz will take much longer than initially anticipated.

The 'end game' in this scenario could be a temporary ceasefire or tacit stand down with maritime escorts and exploratory talks on sanctions. Regime change over time is still possible, driven from the inside, as in East Germany in the late 1980s.

- **Scenario 2** (optimistic alternative): War ends surprisingly earlier than in our new base case scenario and the Strait of Hormuz opens up within a short period.
- **Scenario 3** (longer war): Longer combat action followed by protracted, lower-grade, impulsive confrontation. That would imply a much more uncertain situation in the Strait. In this scenario, the 'end game' could be a hard won ceasefire with third party guarantees (likely involving Russia or China), the possibility of sequenced sanctions relief, and a maritime security regime for Hormuz. However, even then, a durable peace could remain elusive, and the region might settle into a fragile, flare up prone equilibrium.

In short, our base case scenario does not foresee structural damage to an oil or gas production facility. We are facing a delay in energy and other shipments, but not (yet) a structural reduction. Needless to say, if a delay lasts long enough, it can still lead to these structural reductions. Generally speaking and for the time being, this is still mainly a supply-side shock, pushing up inflation and posing new challenges for central banks. As a result, the upward revision to our inflation forecasts is clearly larger than the downward revisions of growth, and they also differ across regions. What these new scenarios mean for our market forecasts can also be found [here](#).

Contact us directly to discuss more details of the underlying scenarios and their implications.

ING Global Forecasts

	1Q26F	2Q26F	2026 3Q26F	4Q26F	2026F	1Q27F	2Q27F	2027 3Q27F	4Q27F	2027F
United States										
GDP (% QoQ, ann)	3.1	2.4	1.8	1.9	2.5	1.9	2.0	2.0	2.0	2.0
CPI headline (% YoY, avg)	2.5	3.8	3.5	3.2	3.3	2.8	2.0	1.9	2.1	2.2
Federal funds (% eop)	3.75	3.75	3.50	3.25	3.25	3.25	3.25	3.25	3.25	3.25
3-month SOFR rate (% eop)	3.70	3.70	3.50	3.30	3.30	3.30	3.30	3.30	3.30	3.30
10-year interest rate (% eop)	4.20	4.30	4.20	4.15	4.15	4.15	4.25	4.25	4.30	4.30
Fiscal balance (% of GDP)					-6					-6.1
Gross public debt / GDP					102.6					104.5
Eurozone										
GDP (% QoQ, ann)	0.7	0.6	1.0	1.5	0.9	1.6	1.6	1.5	1.4	1.4
CPI headline (% YoY, avg)	2.0	3.2	3.1	2.7	2.8	2.6	1.9	1.8	2.0	2.1
ECB Deposit Rate (% eop)	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
3-month interest rate (% eop)	2.00	2.00	2.10	2.10	2.10	2.10	2.10	2.10	2.10	2.10
10-year interest rate (% eop)	3.00	3.00	2.95	2.85	2.85	2.90	3.00	3.00	3.10	3.10
Fiscal balance (% of GDP)					-3.6					-3.5
Gross public debt/GDP					93					93.9
Japan										
GDP (% QoQ, ann)	1.2	0.8	1.2	1.6	0.8	0.4	1.2	0.8	0.8	1.0
CPI headline (% YoY, avg)	1.7	2.4	2.4	2.3	2.2	2.6	1.8	1.8	1.6	2.0
Target rate (% eop)	0.75	1.00	1.00	1.00	1.00	1.25	1.25	1.25	1.50	1.50
3-month interest rate (% eop)	1.20	1.20	1.20	1.35	1.35	1.50	1.50	1.70	1.75	1.75
10-year interest rate (% eop)	2.25	2.50	2.50	2.60	2.60	2.75	2.75	2.85	2.85	2.85
Fiscal balance (% of GDP)					-4.6					-5.0
Gross public debt/GDP					230					228
China										
GDP (% YoY)	4.6	4.6	4.7	4.5	4.6	4.2	4.9	4.4	4.4	4.5
CPI headline (% YoY, avg)	0.9	1.1	1.4	1.3	1.2	1.6	1.1	1.1	1.1	1.3
7-day Reverse Repo Rate (% eop)	1.40	1.40	1.30	1.30	1.30	1.30	1.20	1.20	1.20	1.20
3M SHIBOR (% eop)	1.55	1.50	1.45	1.45	1.45	1.45	1.45	1.45	1.45	1.45
10-year T-bond yield (% eop)	1.85	1.90	1.95	2.00	2.00	2.00	2.05	2.10	2.15	2.15
Fiscal balance (% of GDP)					-5.3					-5.3
Public debt (% of GDP), ind, local govt					145					-
United Kingdom										
GDP (% QoQ, ann)	1.3	0.9	-0.1	0.8	0.6	1.5	1.5	1.5	1.5	1.1
CPI headline (% YoY, avg)	3.0	2.4	3.5	3.4	3.1	3.2	2.9	1.9	2.1	2.5
BoE official bank rate (% eop)	3.75	3.50	3.25	3.25	3.25	3.25	3.25	3.25	3.25	3.25
3-month interest rate (% eop)	3.50	3.20	3.20	3.20	3.20	3.20	3.20	3.20	3.20	3.20
10-year interest rate (% eop)	4.70	4.60	4.50	4.30	4.30	4.30	4.40	4.40	4.40	4.40
Fiscal balance (% of GDP)					3.6					3.1
Public sector net debt (FY, %)					96.1					96.4
EUR/USD (eop)	1.15	1.16	1.18	1.20	1.20	1.20	1.20	1.20	1.20	1.20
USD/JPY (eop)	158	158	155	153	153	150	148	147	145	145
USD/CNY (eop)	6.90	6.87	6.82	6.80	6.80	6.78	6.75	6.75	6.70	6.70
EUR/GBP (eop)	0.87	0.88	0.89	0.90	0.90	0.90	0.90	0.90	0.90	0.90
ICE Brent - US\$/bbl (average)	76	91	85	77	82	72	70	67	63	68
Dutch TTF - EUR/MWh (average)	39	50	38	38	41	34	28	27	30	30

Source: ING forecasts

GDP Forecasts

QoQ% Annualised (avg)										
	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
US	3.1	2.4	1.8	1.9	1.9	2.0	2.0	2.0	2.5	2.0
Japan	1.2	0.8	1.2	1.6	0.4	1.2	0.8	0.8	0.8	1.0
Germany	-0.5	1.1	1.8	2.2	1.8	2.6	2.5	2.5	0.8	2.2
France	0.6	0.4	0.4	0.6	1.2	1.4	1.6	1.2	0.8	1.0
UK	1.3	0.9	-0.1	0.8	1.5	1.5	1.5	1.5	0.6	1.1
Italy	0.3	0.4	0.9	1.3	0.9	0.6	0.8	0.6	0.7	0.8
Canada	1.8	1.5	1.4	1.6	2.2	2.0	2.1	2.0	1.1	1.9
Australia	2.3	2.2	2.3	2.2	2.5	2.5	2.5	2.5	2.3	2.5
Eurozone	0.7	0.6	1.0	1.5	1.6	1.6	1.5	1.4	0.9	1.4
Austria	1.0	0.6	1.2	1.8	1.8	1.6	1.6	1.4	0.8	1.6
Spain	1.9	1.1	2.0	1.5	2.1	2.1	2.0	2.0	2.1	1.9
Netherlands	0.9	0.9	1.5	1.6	1.4	1.5	1.1	1.0	1.4	1.4
Belgium	0.8	0.4	1.2	1.4	1.3	1.3	1.2	1.4	0.8	1.2
Greece	1.6	0.8	1.8	1.7	1.7	2.4	2.2	1.7	1.9	1.8
Portugal	1.7	1.3	1.6	1.7	2.0	2.0	2.0	2.0	2.1	1.9
Switzerland	1.2	0.6	0.8	1.2	1.2	1.6	1.6	1.2	0.5	1.2

Emerging Markets, (YoY% growth)										
	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
Bulgaria	3.1	2.8	2.7	2.5	2.5	2.5	2.5	2.7	2.8	2.5
Croatia	3.5	3.1	3.0	1.7	1.6	1.7	1.8	1.8	2.7	1.7
Czech Republic	2.5	2.8	2.7	2.8	2.8	2.8	2.7	2.7	2.7	2.7
Hungary	1.6	1.4	1.7	2.2	1.9	3.2	3.1	3.6	1.7	3.0
Poland	3.6	3.8	3.7	3.6	3.6	3.4	3.0	3.0	3.7	3.2
Romania	-0.7	-0.8	0.3	2.8	3.1	2.9	2.6	2.7	0.6	2.8
Turkey	3.2	2.8	3.6	3.8	4.4	5.5	5.2	4.9	3.4	5.0
Serbia	3.5	3.0	3.1	2.6	2.8	3.2	3.3	3.6	3.0	3.2
Azerbaijan	3.0	2.0	4.0	1.0	3.0	3.5	2.0	3.5	2.5	3.0
Kazakhstan	4.0	5.0	5.0	6.0	5.0	4.8	4.0	4.0	5.0	4.5
Russia	1.0	1.0	0.5	-0.5	0.0	0.5	1.0	1.5	0.5	0.8
Uzbekistan	6.5	6.3	6.0	7.0	6.5	5.5	6.0	6.0	6.5	6.0
Ukraine	2.1	2.6	2.5	2.8	3.2	3.4	3.5	3.5	2.5	3.4
China	4.6	4.6	4.7	4.5	4.2	4.9	4.4	4.4	4.6	4.5
India	6.5	6.6	6.8	7.0	6.5	6.5	6.5	6.5	6.7	6.5
Indonesia	4.8	4.9	5.0	5.0	5.0	5.0	5.0	5.0	4.9	5.0
Korea	2.4	2.4	1.5	2.3	1.8	1.8	1.9	1.8	2.2	1.8
Philippines	4.0	4.5	6.0	6.2	6.0	6.0	6.0	6.0	5.2	6.0
Singapore	4.5	4.0	3.0	2.0	1.8	1.8	1.8	1.8	3.4	1.8
Taiwan	10.2	7.4	6.9	3.1	4.0	4.6	5.0	5.1	6.7	4.7

Norway: Forecasts are mainland GDP

Source: ING estimates

CPI Forecasts

YoY% (avg)	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
US	2.5	3.8	3.5	3.2	2.8	2.0	1.9	2.1	3.3	2.2
Japan	1.7	2.4	2.4	2.3	2.6	1.8	1.8	1.6	2.2	2.0
Germany	2.2	3.2	3.2	3.0	3.0	2.2	2.1	2.1	2.9	2.3
France	1.1	2.4	2.2	2.3	1.9	1.1	0.8	1.5	2.0	1.3
UK	3.0	2.4	3.5	3.4	3.2	2.9	1.9	2.1	3.1	2.5
Italy	1.5	2.3	2.6	2.6	2.2	1.7	1.6	1.9	2.3	1.8
Canada	2.1	2.8	2.9	2.5	2.1	1.7	1.8	2.0	2.6	1.9
Australia	3.7	3.7	3.2	3.2	3.0	3.0	3.0	3.0	3.5	3.0
Eurozone	2.0	3.2	3.1	2.7	2.6	1.9	1.8	2.0	2.8	2.1
Austria	2.3	2.7	2.6	2.3	2.2	1.9	1.8	2.0	2.5	2.0
Spain	2.6	3.3	2.8	2.6	2.2	2.2	2.1	2.1	2.8	2.1
Netherlands	2.3	2.5	3.0	3.1	2.6	2.3	1.7	1.6	2.7	2.0
Belgium	1.8	3.4	3.3	3.0	2.8	2.0	1.9	1.9	3.0	2.2
Greece	3.1	3.5	3.6	3.1	2.7	2.0	2.2	2.2	3.3	2.3
Portugal	2.2	2.9	2.6	2.4	2.0	2.0	1.9	1.9	2.5	2.0
Switzerland	0.4	1.1	1.0	0.7	0.6	0.6	0.8	0.8	0.8	0.7
Bulgaria	3.3	3.7	2.3	2.5	2.4	2.3	2.5	2.6	3.0	2.5
Croatia	3.8	4.1	3.3	3.2	2.4	2.6	2.7	3.1	3.6	2.7
Czech Republic	1.5	1.8	1.6	2.1	2.5	2.3	2.3	2.3	1.7	2.3
Hungary	1.9	3.4	3.6	3.9	4.0	4.2	3.8	3.4	3.2	3.8
Poland	2.5	3.6	3.7	3.3	3.1	2.2	2.0	2.3	3.2	2.4
Romania	9.7	10.4	6.0	5.4	4.2	4.1	3.9	3.7	7.9	4.0
Turkey	31.0	28.6	26.3	25.5	21.6	20.7	19.0	18.3	28.1	20.3
Serbia	2.6	3.0	2.6	3.4	3.3	3.5	3.8	4.2	3.0	3.8
Azerbaijan	5.8	5.5	4.9	4.6	4.3	4.5	4.7	4.9	5.2	4.6
Kazakhstan	11.8	11.0	10.3	10.4	10.0	9.3	8.6	8.1	10.9	9.0
Russia	5.9	5.4	4.8	5.3	4.5	4.8	5.3	4.7	5.3	4.8
Uzbekistan	7.4	7.1	7.0	7.7	8.0	7.7	7.6	7.6	7.3	7.7
Ukraine	9.2	8.5	7.5	7.0	6.5	6.3	6.1	5.7	8.1	6.2
China	0.9	1.1	1.4	1.3	1.6	1.1	1.1	1.1	1.2	1.3
India	3.0	3.7	4.1	4.3	4.5	4.8	4.8	5.0	3.8	5.0
Indonesia	3.0	2.3	2.5	2.5	2.5	2.5	3.0	3.0	2.6	2.5
Korea	2.1	3.0	2.5	2.1	1.8	1.3	1.9	2.2	2.4	1.8
Philippines	2.5	3.0	3.2	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Singapore	1.6	1.8	1.9	1.8	2.0	2.0	2.0	2.0	1.8	2.0
Taiwan	1.4	2.6	2.5	1.9	1.5	1.3	1.4	1.4	2.1	1.4

*Quarterly forecasts are quarterly average; yearly forecasts are average over the year, HICP for Eurozone economies

Source: ING estimates

Oil and Natural Gas Forecasts (avg)

	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
Brent (\$/bbl)	76	91	85	77	72	70	67	63	82	68
Dutch TTF (EUR/MWh)	39	50	38	38	34	28	27	30	41	30

Source: ING estimates

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Assessing the global economic impact of the Middle East war

The Middle East war remains highly unpredictable, with even seasoned analysts unable to gauge how it might end. Here, we outline what the conflict could mean for the global economy



Escalating US-Iran confrontation raises the risk of regional spillovers with global economic consequences

Needless to say, the war in the Middle East is one of these fast-moving environments, and even the best geopolitical analysts are struggling to predict how it could end. For financial markets, the branching point is simple and more brutal: does this end in days, or does it become a forever war that involves an entire region? Crucial factors here are whether, how and when there could be a regime change in Iran; the duration and intensity of US involvement; and the length and severity of any disruption to traffic through the Strait of Hormuz.

In shaping our view, we are working with a base case scenario that assumes roughly two weeks of ongoing combat, involving not only the US, Israel and Iran but the entire Middle East region. Air traffic will be down and the Strait of Hormuz will be basically blocked. After these two weeks, political instability in Iran will remain, but broader uncertainty will recede and the Strait of Hormuz will gradually reopen. Within four to six weeks, conditions could return close to the pre war environment, albeit with elevated levels of uncertainty. Given that there is no historical evidence of quick and smooth regime change in any country, the risk to our base case scenario is clearly to the downside. Think of 'boots on the ground', much longer-lasting military action or Iranian retaliation via activated sleeper cells or cyberattacks on US targets or even US territory.

Global trade: A supply shock at the worst possible moment

The Iran war is unfolding against a global trading system already strained by Trump's tariff offensive and the lingering fragmentation of supply chains since Covid and the war in Ukraine. The Strait of Hormuz is the single most important chokepoint in global energy trade, and it now sits in an active warzone.

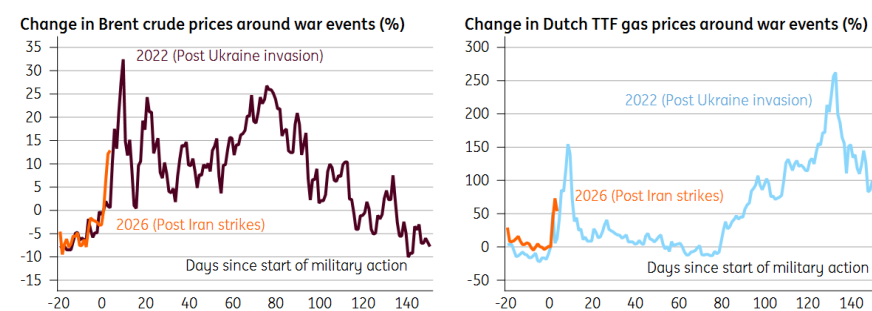
Even without a formal blockade, the commercial consequences are already emerging: insurers are cancelling cover, shipping premiums are spiking, and vessels are re-routing or pausing transits. The knock-on effects extend well beyond energy. Gulf airspace closures are disrupting aviation corridors between Europe and Asia. Houthi reactivation in the Red Sea would close the alternative routing valve that kept goods moving during earlier episodes of Hormuz tension.

The combination of higher energy costs, disrupted logistics, and a generalised confidence shock would constitute a meaningful drag on global trade volumes at precisely the moment the world economy was still digesting the inflationary and growth consequences of the tariff shock. The mother of all bad timings.

How the current energy supply risks compare to 2022

For oil and gas markets, there will likely be some parallels made with 2022 and Russia's invasion of Ukraine. The oil supply at risk from a successful blockade of the Strait of Hormuz is roughly 15-20% of global supply, depending on how much supply the Saudis can divert by pipeline to the Red Sea. This is significantly higher than the 7-8m b/d of Russian oil supply (which is around 7-8% of global supply) that was at risk during the early days of the Russia-Ukraine war, when we saw Brent spike to almost \$140/bbl.

How the reaction in energy markets compares to 2022



However, helping to a certain degree at the moment is the fact that oil inventories are more comfortable than they were in the lead-up to Russia's invasion of Ukraine. OECD stocks are in the region of 200m barrels higher now than prior to the Russia/Ukraine war. Still, a two-week full blockade would essentially see this buffer disappear, leaving significant upside to prices.

For natural gas, as much as 125bcm of LNG flows are at risk, which is around 3% of global natural gas consumption, but 22% of global LNG trade. However, the roughly 15bcm that Oman exports will be at less of a risk, given cargoes don't need to navigate the Strait of Hormuz, but will still be in close proximity to danger.

In the lead-up to Russia's invasion of Ukraine, close to 160bcm of Russian gas (pipeline and LNG to the EU) was at risk. The market is relatively better positioned now, given the build-up of LNG export capacity, largely from the US. Since the beginning of 2025, we have seen around 40bcm of US capacity starting up, while a further 14bcm is set to start this year, and there will be significantly more in the years ahead. However, in the immediate term, capacity additions fall well short of potential Persian Gulf supply losses.

A tighter market would see Asia and Europe competing more aggressively for LNG cargoes, pushing up prices. Price-sensitive buyers in Asia will likely step back from the market, while Europe would likely not make the same mistake as it did in 2022, where buyers bought aggressively regardless of price levels.

In the eurozone – and similarly in the US – the labour market is another important way in which today differs from 2022. In the UK, for example, worker shortages were still widespread in 2022, which amplified the pass through from higher inflation to wage growth. The jobs market is now much cooler, so that mechanism should be far less pronounced. The same likely applies to the eurozone, judging by European Commission surveys showing fewer firms reporting labour shortages as a constraint on production.

At the same time, what is different from a macro perspective compared with 2022 is the fact that in 2022 most major economies looked resilient and governments, which had already introduced fiscal support during the pandemic, announced more measures to prevent too much of a purchasing power loss due to higher energy prices.

The US: Fighting a war that raises domestic prices

For the US, even though trade exposure to the Strait of Hormuz is limited, higher global oil prices would fuel the current cost-of-living crisis. US consumers are already stretched, and gasoline prices are acutely politically sensitive going into midterm territory. Higher oil prices would also complicate the Federal Reserve's future monetary policy path.

A second supply-side inflation shock, while the inflationary impact from tariffs is still unfolding, could make further rate cuts hard to justify, at least in the near term. At the same time, if the conflict drags and uncertainty weighs on business investment and consumer confidence, the growth outlook darkens too.

The one partial offset is that the US itself is a major oil producer; higher oil prices benefit the shale patch and improve the terms of trade for domestic energy, even as they hurt consumers. But that balance is politically awkward to explain and economically insufficient to compensate for the broader damage.

The eurozone: The most exposed major economy

Europe is where the macro consequences hit hardest, and the timing could not be worse. The eurozone was finally emerging from its long period of stagnation, with tentative green shoots of recovery emerging – though recently, these have been undermined by new uncertainty regarding tariffs. Now the region could face an energy shock on top of a trade shock.

Europe imports essentially all of its oil and a significant share of its LNG. A surge in energy prices and potentially even energy supply disruption could bring back memories of the energy cost crisis

from late 2021 to 2023. There are currently two important differences compared with the situation back then: Europe doesn't have to 'derisk' from a single important energy provider, and the oil price crisis comes at the end of the winter, not the start. Whether eurozone governments will be able (and willing) to quickly offset any purchasing power losses via new fiscal support, however, is less clear. Fiscal capacities are more under pressure than in 2022.

The European Central Bank is caught in a genuine dilemma. Services inflation is still sticky, and an oil shock would push headline inflation higher – yet the growth outlook is simultaneously deteriorating under the combined weight of tariffs, uncertainty, and now energy costs. In December, ECB analysis showed that a 14% increase in oil prices and 20% in gas prices would push up inflation by 0.5ppt and could reduce GDP growth by 0.1ppt. However, this would only be the price effect, not the supply chain disruption effect. Given the still relatively fresh memories of the recent inflation surge, the ECB is unlikely to see any new oil price-driven inflation spike as transitory or even deflationary. However, to see a rate hike, the eurozone economy would have to show clear resilience.

Asia: Inflation and trade balances could come under strain

For now, Asia seems to be able to absorb the jump in oil prices, thanks to the low starting points, with inflation broadly in control in most of Asia. However, the severity and persistence of higher prices will ultimately determine the impact. If sustained, Asia is particularly vulnerable to oil price volatility because it relies so heavily on imports; except for Australia, Malaysia and Indonesia, all other economies run deficits in oil and gas trade, leaving them exposed when energy costs rise. If higher prices persist, three factors will shape the impact:

1. Heavy dependence on Middle Eastern oil: A significant share of Asia's crude supply comes from the Persian Gulf. Japan and the Philippines rely on the region for almost 90% of their oil needs, while China and India import roughly 38% and 46%, respectively. Any disruption in the Strait of Hormuz – a critical shipping lane – would restrict supply, potentially causing shortages that slow business activity and put pressure on manufacturing across Asia.
2. Trade balances under strain: Even without a physical supply disruption, higher global oil prices worsen trade balances and add to inflation pressures. Thailand, Korea, Vietnam, Taiwan, and the Philippines are the most exposed. A mere 10% rise in oil prices can deteriorate current account balances by 40-60 basis points. Prolonged increases would only deepen these deficits.
3. Strong inflation pass through: Because many emerging Asian economies have a relatively high weight of energy in their consumer inflation baskets, rising oil prices feed quickly into headline inflation. On average, a 10% increase in oil prices raises CPI inflation by about 0.2 percentage points.

Our base case had headline inflation across most of Asia rising in 2026 but still staying within most central bank targets. But a price shock of this magnitude – if it lasts – would likely push inflation above target ranges and increase the pressure on central banks to tighten policy sooner rather than later.

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